Stuxnet

Stuxnet is a malicious computer worm first uncovered on 17 June 2010[2] and thought to have been in development since at least 2005. Stuxnet targets supervisory control and data acquisition (SCADA) systems and is believed to be responsible for causing substantial damage to the Iran nuclear program after it was first installed on a computer at the Natanz Nuclear Facility in 2009.[3][4] Although neither the United States nor Israel has openly admitted responsibility, multiple independent news organizations claim Stuxnet to be a cyberweapon built jointly by the two countries in a collaborative effort known as Operation Olympic Games.[5][6][7] The program, started during the Bush administration, was rapidly expanded within the first months of Barack Obama's presidency.[8]

Stuxnet specifically targets programmable logic controllers (PLCs), which allow the automation of electromechanical processes such as those used to control machinery and industrial processes including gas centrifuges for separating nuclear material. Exploiting four zero-day flaws in the systems,[9] Stuxnet functions by targeting machines using the Microsoft Windows operating system and networks, then seeking out Siemens Step7 software. Stuxnet reportedly compromised Iranian PLCs, collecting information on industrial systems and causing the fast-spinning centrifuges to tear themselves apart.[3] Stuxnet's design and architecture are not domain-specific and it could be tailored as a platform for attacking modern SCADA and PLC systems (e.g., in factory assembly lines or power plants), most of which are in Europe, Japan and the United States.[10] Stuxnet reportedly destroyed almost one-fifth of Iran's nuclear centrifuges.[11] Targeting industrial control systems, the worm infected over 200,000 computers and caused 1,000 machines to physically degrade.[12]

Stuxnet	
Malware details	
Technical name	**As Stuxnet** ▪ By Microsoft Worm:Win32/Stuxnet.[Letter] TrojanDropper:Win32/Stuxnet ▪ By Symantec W32.Stuxnet W32.Stuxnet!lnk ▪ By Sophos Troj/Stuxnet-[Letter] Trojan-Dropper.Win32.Stuxnet.[Letter] Worm.Win32.Stuxnet.[Letter] TR/Drop.Stuxnet.[Letter].[Number] ▪ By Kaspersky Worm.Win32.Stuxnet ▪ By F-Secure Trojan-Dropper:W32/Stuxnet Rootkit:W32/Stuxnet ▪ By Trend Micro RTKT_STUXNET.[Letter] LNK_STUXNET.[Letter] WORM_STUXNET.[Letter]
Type	Dropper
Classification	Computer worm
Origin	United States

Stuxnet has three modules: a worm that executes all routines related to the main payload of the attack, a link file that automatically executes the propagated copies of the worm and a rootkit component responsible for hiding all malicious files and processes to prevent detection of Stuxnet.[13] It is typically introduced to the target environment via an infected USB flash drive, thus crossing any air gap. The worm then propagates across the network, scanning for Siemens Step7 software on computers controlling a PLC. In the absence of either criterion, Stuxnet becomes dormant inside the computer. If both the conditions are fulfilled, Stuxnet introduces the infected rootkit onto the PLC and Step7 software, modifying the code and giving unexpected commands to the PLC while returning a loop of normal operation system values back to the users.[14][15]

Author	Equation Group
	Technical details
Platforms	Windows 2000 Windows XP Windows Server 2003 Windows Vista Windows Server 2008 Windows 7 Windows Server 2008 R2 Source:[1]
Size	~0.5MB
Written in	C, C++ and others

Discovery

Stuxnet, discovered by Sergey Ulasen from a Belarusian antivirus company VirusBlokAda, initially spread via Microsoft Windows, and targeted Siemens industrial control systems. While it is not the first time that hackers have targeted industrial systems,[16] nor the first publicly known intentional act of cyberwarfare to be implemented, it is the first discovered malware that spies on and subverts industrial systems,[17] and the first to include a programmable logic controller (PLC) rootkit.[18][19]

The worm initially spreads indiscriminately, but includes a highly specialized malware payload that is designed to target only Siemens supervisory control and data acquisition (SCADA) systems that are configured to control and monitor specific industrial processes.[20][21] Stuxnet infects PLCs by subverting the Step-7 software application that is used to reprogram these devices.[22][23]

Different variants of Stuxnet targeted five Iranian organizations,[24] with the probable target widely suspected to be uranium enrichment infrastructure in Iran;[23][25][26] Symantec noted in August 2010 that 60 percent of the infected computers worldwide were in Iran.[27] Siemens stated that the worm caused no damage to its customers,[17] but the Iran nuclear program, which uses embargoed Siemens equipment procured secretly, was damaged by Stuxnet.[28][29][30] Kaspersky Lab concluded that the sophisticated attack could only have been conducted "with nation-state support".[31] F-Secure's chief researcher Mikko Hyppönen, when asked if possible nation-state support were involved, agreed: "That's what it would look like, yes."[32]

In May 2011, the PBS program *Need To Know* cited a statement by Gary Samore, White House Coordinator for Arms Control and Weapons of Mass Destruction, in which he said "we're glad they [the Iranians] are having trouble with their centrifuge machine and that we – the U.S. and its allies – are doing everything we can to make sure that we complicate matters for them", offering "winking

acknowledgement" of United States involvement in Stuxnet.[33] According to *The Daily Telegraph*, a showreel that was played at a retirement party for the head of the Israel Defense Forces (IDF), Gabi Ashkenazi, included references to Stuxnet as one of his operational successes as the IDF chief of staff.[34]

On 1 June 2012, an article in *The New York Times* reported that Stuxnet was part of a US and Israeli intelligence operation named *Operation Olympic Games*, devised by the NSA under President George W. Bush and executed under President Barack Obama.[35]

On 24 July 2012, an article by Chris Matyszczyk from CNET[36] reported that the Atomic Energy Organization of Iran e-mailed F-Secure's chief research officer Mikko Hyppönen to report a new instance of malware.

On 25 December 2012, an Iranian semi-official news agency announced there was a cyberattack by Stuxnet, this time on the industries in the southern area of the country. The malware targeted a power plant and some other industries in Hormozgan province in 2012.[37]

According to Eugene Kaspersky, the worm also infected a nuclear power plant in Russia. Kaspersky noted, however, that since the power plant is not connected to the public Internet, the system should remain safe.[38]

History

The worm was first identified by the security company VirusBlokAda in mid-June 2010.[22] Journalist Brian Krebs's blog post on 15 July 2010 was the first widely read report on the worm.[39][40] The original name given by VirusBlokAda was "Rootkit.Tmphider;"[41] Symantec, however, called it "W32.Temphid", later changing it to "W32.Stuxnet".[42] Its current name is derived from a combination of keywords found in the software (".stub" and "mrxnet.sys").[43][44] The timing of the discovery has been attributed to the virus accidentally spreading beyond its intended target due to a programming error introduced in an update. This may have caused the worm to spread to an engineer's computer connected to the centrifuges, further propagating when the engineer later connected to the internet at home.[35]

Kaspersky Lab experts initially estimated that Stuxnet began spreading around March or April 2010,[45] but the first variant of the worm appeared in June 2009.[22] On 15 July 2010, the day the worm's existence became widely known, a distributed denial-of-service attack targeted the servers of two leading mailing lists on industrial-systems security. This attack, from an unknown source but possibly related to Stuxnet, disabled one of the lists, interrupting a key information source for power plants and factories.[40] Separately, researchers at Symantec uncovered a version of the Stuxnet computer virus that was used to attack Iran's nuclear program in November 2007, with evidence indicating it was under development as early as 2005, when Iran was still setting up its uranium enrichment facility.[46]

The second variant, with substantial improvements, appeared in March 2010, reportedly due to concerns that Stuxnet was not spreading fast enough. A third variant, with minor improvements, followed in April 2010.[40] The worm contains a component with a build timestamp from 3 February 2010.[47] On 25 November 2010, Sky News in the United Kingdom reported receiving information from an anonymous source at an unidentified IT security organization claiming that Stuxnet, or a variation of the worm, had been traded on the black market.[48]

In 2015, Kaspersky Lab reported that the Equation Group had used two of the same zero-day attacks prior to their use in Stuxnet, in another malware called fanny.bmp.[49][50] Kaspersky Lab noted that "the similar type of usage of both exploits together in different computer worms, at around the same time, indicates that the Equation Group and the Stuxnet developers are either the same or working closely together".[51]

In 2019, *Chronicle* researchers Juan Andres Guerrero-Saade and Silas Cutler presented findings indicating that at least four distinct threat actor malware platforms collaborated in developing the different versions of Stuxnet.[52][53] The collaboration was referred to as 'GOSSIP GIRL', a name derived from a threat group mentioned in classified CSE slides that included Flame.[54] GOSSIP GIRL is described as a cooperative umbrella encompassing the Equation Group, Flame, Duqu, and Flowershop (also known as 'Cheshire Cat').[55][56][57]

In 2020, researcher Facundo Muñoz presented findings suggesting that Equation Group may have collaborated with Stuxnet developers in 2009 by providing at least one zero-day exploit,[58] and one exploit from 2008[59] that was actively used by the Conficker computer worm and Chinese hackers.[60] In 2017, a group of hackers known as The Shadow Brokers leaked a collection of tools attributed to Equation Group, including new versions of both exploits compiled in 2010. Analysis of the leaked data indicated significant code overlaps, as both Stuxnet's exploits and Equation Group's exploits were developed using a set of libraries called the "Exploit Development Framework", also leaked by The Shadow Brokers.

Affected countries

A study of the spread of Stuxnet by Symantec showed that the main affected countries in the early days of the infection were Iran, Indonesia and India:[61]

Country	Share of infected computers
Iran	58.9%
Indonesia	18.2%
India	8.3%
Azerbaijan	2.6%
United States	1.6%
Pakistan	1.3%
Other countries	9.2%

Iran was reported to have fortified its cyberwar abilities following the Stuxnet attack, and has been suspected of retaliatory attacks.[62][63] These include attacks against United States banks in the Operation Ababil campaign of 2012-2013,[64] the 2012 Shamoon attack against oil giant Saudi Aramco,[65][66] and the 2014 strike against Las Vegas Sands Corporation.[67][68]

Operation

Unlike most malware, Stuxnet does little harm to computers and networks that do not meet specific configuration requirements; "The attackers took great care to make sure that only their designated targets were hit ... It was a marksman's job."[69] While the worm is promiscuous, it makes itself inert if Siemens software is not found on infected computers, and contains safeguards to prevent each infected computer from spreading the worm to more than three others, and to erase itself on 24 June 2012.[40]

For its targets, Stuxnet contains, among other things, code for a man-in-the-middle attack that fakes industrial process control sensor signals so an infected system does not shut down due to detected abnormal behavior.[40][69][70] Such complexity is unusual for malware. The worm consists of a layered attack against three different systems:

1. The Windows operating system,
2. Siemens PCS 7, WinCC and STEP7 industrial software applications that run on Windows and
3. One or more Siemens S7 PLCs.

Windows infection

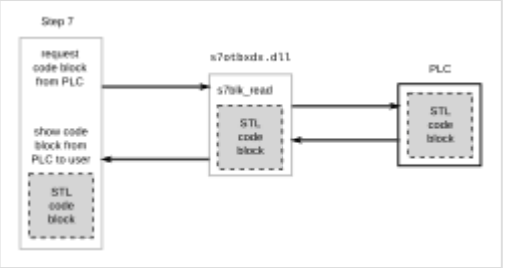
Stuxnet attacked Windows systems using an unprecedented four zero-day attacks (plus the CPLINK vulnerability and a vulnerability used by the Conficker worm[71]). It is initially spread using infected removable drives such as USB flash drives,[23][47] which contain Windows shortcut files to initiate executable code.[72] The worm then uses other exploits and techniques such as peer-to-peer remote procedure call (RPC) to infect and update other computers inside private networks that are not directly connected to the Internet.[73][74][75] The number of zero-day exploits used is unusual, as they are highly valued and malware creators do not typically make use of (and thus simultaneously make visible) four different zero-day exploits in the same worm.[25] Amongst these exploits were remote code execution on a computer with Printer Sharing enabled,[76] and the LNK/PIF vulnerability,[77] in which file execution is accomplished when an icon is viewed in Windows Explorer, negating the need for user interaction.[78] Stuxnet is unusually large at half a megabyte in size,[73] and written in several different programming languages (including C and C++) which is also irregular for malware.[17][22][70] The Windows component of the malware is promiscuous in that it spreads relatively quickly and indiscriminately.[47]

The malware has both user mode and kernel mode rootkit ability under Windows,[75] and its device drivers have been digitally signed with the private keys of two public key certificates that were stolen from separate well-known companies, JMicron and Realtek, both located at Hsinchu Science Park in Taiwan.[47][73] The driver signing helped it install kernel mode rootkit drivers successfully without users being notified, and thus it remained undetected for a relatively long period of time.[79] Both compromised certificates have been revoked by Verisign.

Two websites in Denmark and Malaysia were configured as command and control servers for the malware, allowing it to be updated, and for industrial espionage to be conducted by uploading information. Both of these domain names have subsequently been redirected by their DNS service provider to Dynadot as part of a global effort to disable the malware.[75][40]

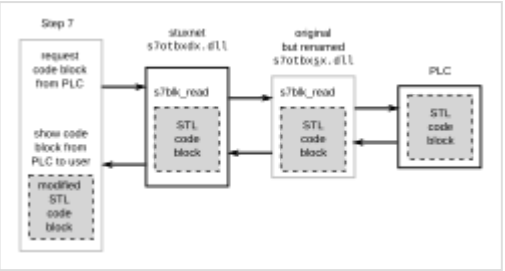
Step 7 software infection

According to researcher Ralph Langner,[80][81] once installed on a Windows system, Stuxnet infects project files belonging to Siemens' WinCC/PCS 7 SCADA control software[82] (Step 7), and subverts a key communication library of WinCC called `s7otbxdx.dll`. Doing so intercepts communications between the WinCC software running under Windows and the target Siemens PLC devices, when the two are connected via a data cable. The malware is able to modify the code on PLC devices unnoticed, and subsequently to mask its presence from WinCC if the control software attempts to read an infected block of memory from the PLC system.[75]



Overview of normal communications between Step 7 and a Siemens PLC

The malware also used a zero-day exploit in the WinCC/SCADA database software in the form of a hard-coded database password.[83]



Overview of Stuxnet hijacking communication between Step 7 software and a Siemens PLC

PLC infection

Siemens Simatic S7-300 PLC CPU with three I/O modules attached

Stuxnet's payload targets only those SCADA configurations that meet criteria that it is programmed to identify.[40]

Stuxnet requires specific subordinate system to be attached to the targeted Siemens S7-300 controller system: variable-frequency drives (frequency converter drives) and its associated modules. It only attacks those PLC systems with variable-frequency drives from two specific vendors: Vacon based in Finland and Fararo Paya based in Iran.[84] Furthermore, it monitors the frequency of the attached motors, and only attacks systems that spin between 807 Hz and 1,210 Hz. This is a much higher frequency than motors typically operate at in most industrial applications, with the notable exception of gas centrifuges.[84] Stuxnet installs malware into memory block DB890 of the PLC that monitors the Profibus messaging bus of the system.[75] When certain criteria are met, it periodically modifies the frequency to 1,410 Hz and then to 2 Hz and then to 1,064 Hz, and thus affects the operation of the connected motors by changing their rotational speed.[84] It also installs a rootkit – the first such documented case on this platform – that hides the malware on the system and masks the changes in rotational speed from monitoring systems.

Removal

Siemens has released a detection and removal tool for Stuxnet. Siemens recommends contacting customer support if an infection is detected and advises installing Microsoft updates for security vulnerabilities and prohibiting the use of third-party USB flash drives.[85] Siemens also advises immediately upgrading

password access codes.[86]

The worm's ability to reprogram external PLCs may complicate the removal procedure. Symantec's Liam O'Murchu warns that fixing Windows systems may not fully solve the infection; a thorough audit of PLCs may be necessary. Despite speculation that incorrect removal of the worm could cause damage,[17] Siemens reports that in the first four months since discovery, the malware was successfully removed from the systems of 22 customers without any adverse effects.[85][87]

Control system security

Prevention of control system security incidents,[88] such as from viral infections like Stuxnet, is a topic that is being addressed in both the public and the private sector.

The US Department of Homeland Security National Cyber Security Division (NCSD) operates the Control System Security Program (CSSP).[89] The program operates a specialized computer emergency response team called the Industrial Control Systems Cyber Emergency Response Team (ICS-CERT), conducts a biannual conference (ICSJWG), provides training, publishes recommended practices, and provides a self-assessment tool. As part of a Department of Homeland Security plan to improve American computer security, in 2008 it and the Idaho National Laboratory (INL) worked with Siemens to identify security holes in the company's widely used Process Control System 7 (PCS 7) and its software Step 7. In July 2008, INL and Siemens publicly announced flaws in the control system at a Chicago conference; Stuxnet exploited these holes in 2009.[69]

Several industry organizations[90][91] and professional societies[92][93] have published standards and best practice guidelines providing direction and guidance for control system end-users on how to establish a control system security management program. The basic premise that all of these documents share is that prevention requires a multi-layered approach, often termed *defense in depth*.[94] The layers include policies and procedures, awareness and training, network segmentation, access control measures, physical security measures, system hardening, e.g., patch management, and system monitoring, anti-virus and intrusion prevention system (IPS). The standards and best practices also all recommend starting with a risk analysis and a control system security assessment.[95][96]

Target and origin

Stuxnet may be the largest and costliest development effort in malware history.[40] Developing its abilities would have required a team of capable programmers, in-depth knowledge of industrial processes, and an interest in attacking industrial infrastructure.[17][22] Eric Byres, who has years of experience maintaining and troubleshooting Siemens systems, told *Wired* that writing the code would have taken many man-months, if not man-years.[73] Symantec estimates that the group developing Stuxnet would have consisted of between five and thirty people, and would have taken six months to prepare.[97][40] *The Guardian*, the BBC and *The New York Times* all claimed that (unnamed) experts studying Stuxnet believe the complexity of the code indicates that only a nation-state would have the abilities to produce it.[25][97][98] The self-destruct and other safeguards within the code implied that a Western government was responsible, or at least is responsible for its development.[40] However, software security expert

Bruce Schneier initially condemned the 2010 news coverage of Stuxnet as hype, stating that it was almost entirely based on speculation.[99] But after subsequent research, Schneier stated in 2012 that "we can now conclusively link Stuxnet to the centrifuge structure at the Natanz nuclear enrichment lab in Iran".[100]

In late December 2008, Dutch engineer Erik van Sabben travelled to Iran, allegedly to infiltrate the Natanz nuclear facility on behalf of Dutch intelligence and install equipment infected with Stuxnet.[101][102] He died two weeks after the Stuxnet attack at age 36 in an apparent single-vehicle motorcycle accident in Dubai.[103]

Iran as a target

Ralph Langner, the researcher who identified that Stuxnet infected PLCs,[23] first speculated publicly in September 2010 that the malware was of Israeli origin, and that it targeted Iranian nuclear facilities.[104] However Langner more recently, at a TED conference, recorded in February 2011, stated that "My opinion is that the Mossad is involved, but that the leading force is not Israel. The leading force behind Stuxnet is the cyber superpower – there is only one; and that's the United States."[105] Kevin Hogan, Senior Director of Security Response at Symantec, reported that most infected systems were in Iran (about 60%),[106] which has led to speculation that it may have been deliberately targeting "high-value infrastructure" in Iran[25] including either the Bushehr Nuclear Power Plant or the Natanz nuclear facility.[73][107][108] Langner called the malware "a one-shot weapon" and said that the intended target was probably hit,[109] although he admitted this was speculation.[73] Another German researcher and spokesman of the German-based Chaos Computer Club, Frank Rieger, was the first to speculate that Natanz was the target.[40]

Natanz nuclear facilities

Anti-aircraft guns guarding Natanz Nuclear Facility

According to the Israeli newspaper *Haaretz*, in September 2010 experts on Iran and computer security specialists were increasingly convinced that Stuxnet was meant "to sabotage the uranium enrichment facility at Natanz – where the centrifuge operational capacity had dropped over the past year by 30 percent".[110] On 23 November 2010 it was announced that uranium enrichment at Natanz had ceased several times because of a series of major technical problems.[111] A "serious nuclear accident" (supposedly the shutdown of some of its centrifuges[112]) occurred at the site in the first half of 2009, which is speculated to have forced Gholam Reza Aghazadeh, the head of the Atomic Energy Organization of Iran (AEOI), to resign.[113] Statistics published by the Federation of American Scientists (FAS) show that the number of enrichment centrifuges operational in Iran mysteriously declined from about 4,700 to about 3,900 beginning around the time the nuclear incident WikiLeaks mentioned would have occurred.[114] The Institute for Science and International Security (ISIS) suggests, in a report published in December 2010, that Stuxnet is a reasonable explanation for the apparent damage[115] at Natanz, and may have destroyed up to 1,000 centrifuges (10 percent) sometime between November 2009 and late January 2010. The authors conclude:

> The attacks seem designed to force a change in the centrifuge's rotor speed, first raising the speed and then lowering it, likely with the intention of inducing excessive vibrations or distortions that would destroy the centrifuge. If its goal was to quickly destroy all the centrifuges in the FEP [Fuel Enrichment Plant], Stuxnet failed. But if the goal was to destroy a more limited number of centrifuges and set back Iran's progress in operating the FEP, while making detection difficult, it may have succeeded, at least temporarily.[115]

The Institute for Science and International Security (ISIS) report further notes that Iranian authorities have attempted to conceal the breakdown by installing new centrifuges on a large scale.[115][116]

The worm worked by first causing an infected Iranian IR-1 centrifuge to increase from its normal operating speed of 1,064 hertz to 1,410 hertz for 15 minutes before returning to its normal frequency. Twenty-seven days later, the worm went back into action, slowing the infected centrifuges down to a few hundred hertz for a full 50 minutes. The stresses from the excessive, then slower, speeds caused the aluminium centrifugal tubes to expand, often forcing parts of the centrifuges into sufficient contact with each other to destroy the machine.[117]

According to *The Washington Post*, International Atomic Energy Agency (IAEA) cameras installed in the Natanz facility recorded the sudden dismantling and removal of approximately 900–1,000 centrifuges during the time the Stuxnet worm was reportedly active at the plant. Iranian technicians, however, were able to quickly replace the centrifuges and the report concluded that uranium enrichment was likely only briefly disrupted.[118]

On 15 February 2011, the Institute for Science and International Security released a report concluding that:

> Assuming Iran exercises caution, Stuxnet is unlikely to destroy more centrifuges at the Natanz plant. Iran likely cleaned the malware from its control systems. To prevent re-infection, Iran will have to exercise special caution since so many computers in Iran contain Stuxnet. Although Stuxnet appears to be designed to destroy centrifuges at the Natanz facility, destruction was by no means total. Moreover, Stuxnet did not lower the production of low enriched uranium (LEU) during 2010. LEU quantities could have certainly been greater, and Stuxnet could be an important part of the reason why they did not increase significantly. Nonetheless, there remain important questions about why Stuxnet destroyed only 1,000 centrifuges. One observation is that it may be harder to destroy centrifuges by use of cyber attacks than often believed.[119]

Iranian reaction

The Associated Press reported that the semi-official Iranian Students News Agency released a statement on 24 September 2010 stating that experts from the Atomic Energy Organization of Iran met in the previous week to discuss how Stuxnet could be removed from their systems.[21] According to analysts, such as David Albright, Western intelligence agencies had been attempting to sabotage the Iranian nuclear program for some time.[120][121]

The head of the Bushehr Nuclear Power Plant told Reuters that only the personal computers of staff at the plant had been infected by Stuxnet and the state-run newspaper *Iran Daily* quoted Reza Taghipour, Iran's telecommunications minister, as saying that it had not caused "serious damage to government systems".[98] The Director of Information Technology Council at the Iranian Ministry of Industries and Mines, Mahmud Liaii, has said that: "An electronic war has been launched against Iran ... This computer worm is designed to transfer data about production lines from our industrial plants to locations outside Iran."[122]

In response to the infection, Iran assembled a team to combat it. With more than 30,000 IP addresses affected in Iran, an official said that the infection was fast spreading in Iran and the problem had been compounded by the ability of Stuxnet to mutate. Iran had set up its own systems to clean up infections and had advised against using the Siemens SCADA antivirus since it is suspected that the antivirus contains embedded code which updates Stuxnet instead of removing it.[123][124][125][126]

According to Hamid Alipour, deputy head of Iran's government Information Technology Company, "The attack is still ongoing and new versions of this virus are spreading." He reported that his company had begun the cleanup process at Iran's "sensitive centres and organizations".[124] "We had anticipated that we could root out the virus within one to two months, but the virus is not stable, and since we started the cleanup process three new versions of it have been spreading", he told the Islamic Republic News Agency on 27 September 2010.[126]

On 29 November 2010, Iranian president Mahmoud Ahmadinejad stated for the first time that a computer virus had caused problems with the controller handling the centrifuges at its Natanz facilities. According to Reuters, he told reporters at a news conference in Tehran: "They succeeded in creating problems for a limited number of our centrifuges with the software they had installed in electronic parts."[127][128]

On the same day two Iranian nuclear scientists were targeted in separate, but nearly simultaneous car bomb attacks near Shahid Beheshti University in Tehran. Majid Shahriari, a quantum physicist, was killed. Fereydoon Abbasi, a high-ranking official at the Ministry of Defense was seriously wounded. *Wired* speculated that the assassinations could indicate that whoever was behind Stuxnet felt that it was not sufficient to stop the nuclear program.[129] That same *Wired* article suggested the Iranian government could have been behind the assassinations.[129] In January 2010, another Iranian nuclear scientist, a physics professor at Tehran University, was killed in a similar bomb explosion.[129] On 11 January 2012, a director of the Natanz nuclear enrichment facility, Mostafa Ahmadi Roshan, was killed in an attack quite similar to the one that killed Shahriari.[130]

An analysis by the FAS demonstrates that Iran's enrichment capacity grew during 2010. The study indicated that Iran's centrifuges appeared to be performing 60% better than in the previous year, which would significantly reduce Tehran's time to produce bomb-grade uranium. The FAS report was reviewed by an official with the IAEA who affirmed the study.[131][132][133]

European and US officials, along with private experts, told Reuters that Iranian engineers were successful in neutralizing and purging Stuxnet from their country's nuclear machinery.[134]

Given the growth in Iranian enrichment ability in 2010, the country may have intentionally put out misinformation to cause Stuxnet's creators to believe that the worm was more successful in disabling the Iranian nuclear program than it actually was.[40]

Israel

Israel, through Unit 8200,[135][136] has been speculated to be the country behind Stuxnet in multiple media reports[97][112][137] and by experts such as Richard A. Falkenrath, former Senior Director for Policy and Plans within the US Office of Homeland Security.[138][98] Yossi Melman, who covers intelligence for Israeli newspaper *Haaretz* and wrote a book about Israeli intelligence, also suspected that Israel was involved, noting that Meir Dagan, the former (up until 2011) head of the national intelligence agency Mossad, had his term extended in 2009 because he was said to be involved in important projects. Additionally, in 2010 Israel grew to expect that Iran would have a nuclear weapon in 2014 or 2015 – at least three years later than earlier estimates – without the need for an Israeli military attack on Iranian nuclear facilities; "They seem to know something, that they have more time than originally thought", he added.[29][69] Israel has not publicly commented on the Stuxnet attack but in 2010 confirmed that cyberwarfare was now among the pillars of its defense doctrine, with a military intelligence unit set up to pursue both defensive and offensive options.[139][140][141] When questioned whether Israel was behind the virus in the fall of 2010, some Israeli officials broke into "wide smiles", fueling speculation that the government of Israel was involved with its genesis.[142] American presidential advisor Gary Samore also smiled when Stuxnet was mentioned,[69] although American officials have suggested that the virus originated abroad.[142] According to *The Telegraph*, Israeli newspaper *Haaretz* reported that a video celebrating operational successes of Gabi Ashkenazi, retiring Israel Defense Forces (IDF) Chief of Staff, was shown at his retirement party and included references to Stuxnet, thus strengthening claims that Israel's security forces were responsible.[143]

In 2009, a year before Stuxnet was discovered, Scott Borg of the United States Cyber-Consequences Unit (US-CCU)[144] suggested that Israel may prefer to mount a cyberattack rather than a military strike on Iran's nuclear facilities.[121] In late 2010 Borg stated: "Israel certainly has the ability to create Stuxnet and there is little downside to such an attack because it would be virtually impossible to prove who did it. So a tool like Stuxnet is Israel's obvious weapon of choice."[145] Iran uses P-1 centrifuges at Natanz, the design for which A. Q. Khan stole in 1976 and took to Pakistan. His black market nuclear-proliferation network sold P-1s to, among other customers, Iran. Experts believe that Israel also somehow acquired P-1s and tested Stuxnet on the centrifuges, installed at the Dimona facility that is part of its own nuclear program.[69] The equipment may be from the United States, which received P-1s from Libya's former nuclear program.[146][69]

Some have also cited several clues in the code such as a concealed reference to the word *MYRTUS*, believed to refer to the Latin name *myrtus* of the Myrtle tree, which in Hebrew is called *hadassah*. Hadassah was the birth name of the former Jewish queen of Persia, Queen Esther.[147][148] However, it may be that the "MYRTUS" reference is simply a misinterpreted reference to SCADA components known as *RTUs* (Remote Terminal Units) and that this reference is actually "My RTUs"–a management feature of SCADA.[149] Also, the number 19790509 appears once in the code and may refer to the date *1979 May 09*, the day Habib Elghanian, a Persian Jew, was executed in Tehran.[75][150][151] Another date that appears in the code is "24 September 2007", the day that Iran's president Mahmoud Ahmadinejad spoke at Columbia University and made comments questioning the validity of the Holocaust.[40] Such data is not conclusive, since, as noted by Symantec, "attackers would have the natural desire to implicate another party".[75]

United States

There has also been several reports on the involvement of the United States and its collaboration with Israel,[152][153] with one report stating that "there is vanishingly little doubt that [it] played a role in creating the worm".[40] It has been reported that the United States, under one of its most secret programs, initiated by the Bush administration and accelerated by the Obama administration,[154] has sought to destroy Iran's nuclear program by novel methods such as undermining Iranian computer systems. A leaked diplomatic cable showed how the United States was advised to target Iran's nuclear abilities through 'covert sabotage'.[155] An article in *The New York Times* in January 2009 credited a then-unspecified program with preventing an Israeli military attack on Iran where some of the efforts focused on ways to destabilize the centrifuges.[156] A *Wired* article claimed that Stuxnet "is believed to have been created by the United States".[157] Dutch historian Peter Koop speculated that the Tailored Access Operations could have developed Stuxnet, possibly in collaboration with Israel.[158]

The fact that John Bumgarner, a former intelligence officer and member of the United States Cyber-Consequences Unit (US-CCU), published an article prior to Stuxnet being discovered or deciphered, that outlined a strategic cyber strike on centrifuges[159] and suggests that cyber attacks are permissible against nation states which are operating uranium enrichment programs that violate international treaties gives some credibility to these claims. Bumgarner pointed out that the centrifuges used to process fuel for nuclear weapons are a key target for *cybertage* operations and that they can be made to destroy themselves by manipulating their rotational speeds.[160]

In a March 2012 interview with *60 Minutes*, retired US Air Force General Michael Hayden – who served as director of both the Central Intelligence Agency and National Security Agency – while denying knowledge of who created Stuxnet said that he believed it had been "a good idea" but that it carried a downside in that it had legitimized the use of sophisticated cyber weapons designed to cause physical damage. Hayden said: "There are those out there who can take a look at this ... and maybe even attempt to turn it to their own purposes". In the same report, Sean McGurk, a former cybersecurity official at the Department of Homeland Security noted that the Stuxnet source code could now be downloaded online and modified to be directed at new target systems. Speaking of the Stuxnet creators, he said: "They opened the box. They demonstrated the capability ... It's not something that can be put back."[161]

Joint effort and other states and targets

In April 2011, Iranian government official Gholam Reza Jalali stated that an investigation had concluded that the United States and Israel were behind the Stuxnet attack.[162] Frank Rieger stated that three European countries' intelligence agencies agreed that Stuxnet was a joint United States-Israel effort. The code for the Windows injector and the PLC payload differ in style, likely implying collaboration. Other experts believe that a US-Israel cooperation is unlikely because "the level of trust between the two countries' intelligence and military establishments is not high".[40]

A *Wired* magazine article about US General Keith B. Alexander stated: "And he and his cyber warriors have already launched their first attack. The cyber weapon that came to be known as Stuxnet was created and built by the NSA in partnership with the CIA and Israeli intelligence in the mid-2000s."[163]

China,[164] Jordan, and France are other possibilities, and Siemens may have also participated.[40][152] Langner speculated that the infection may have spread from USB drives belonging to Russian contractors since the Iranian targets were not accessible via the Internet.[23][165] In 2019, it was reported that an

Iranian mole working for Dutch intelligence at the behest of Israel and the CIA inserted the Stuxnet virus with a USB flash drive or convinced another person working at the Natanz facility to do so.[166][167]

Sandro Gaycken from the Free University Berlin argued that the attack on Iran was a ruse to distract from Stuxnet's real purpose. According to him, its broad dissemination in more than 100,000 industrial plants worldwide suggests a field test of a cyber weapon in different security cultures, testing their preparedness, resilience, and reactions, all highly valuable information for a cyberwar unit.[168]

The United Kingdom has denied involvement in the worm's creation.[169]

In July 2013, Edward Snowden claimed that Stuxnet was cooperatively developed by the United States and Israel.[170]

Deployment in North Korea

According to a report by Reuters, the NSA also tried to sabotage North Korea's nuclear program using a version of Stuxnet. The operation was reportedly launched in tandem with the attack that targeted Iranian centrifuges in 2009–10. The North Korean nuclear program shares a number of similarities with the Iranian, both having been developed with technology transferred by Pakistani nuclear scientist A.Q. Khan. The effort failed, however, because North Korea's extreme secrecy and isolation made it impossible to introduce Stuxnet into the nuclear facility.[171]

Stuxnet 2.0 cyberattack

In 2018, Gholamreza Jalali, Iran's chief of the National Organization for Passive Defense, claimed that his country fended off a Stuxnet-like attack targeting the country's telecom infrastructure. Iran's Telecommunications minister, Mohammad-Javad Azari Jahromi has since accused Israel of orchestrating the attack. Iran plans to sue Israel through the International Court of Justice (ICJ) and is also willing to launch a retaliation attack if Israel does not desist.[172]

Related malware

"Stuxnet's Secret Twin"

A November 2013 article[173] in Foreign Policy magazine claims existence of an earlier, much more sophisticated attack on the centrifuge complex at Natanz, focused on increasing centrifuge failure rate over a long time period by stealthily inducing uranium hexafluoride gas overpressure incidents. This malware was capable of spreading only by being physically installed, probably by previously contaminated field equipment used by contractors working on Siemens control systems within the complex. It is not clear whether this attack attempt was successful, but follow-up by a different, simpler, and more conventional attack is indicative that it was not.

Duqu

On 1 September 2011, a new worm was found, thought to be related to Stuxnet. The Laboratory of Cryptography and System Security (CrySyS) of the Budapest University of Technology and Economics analyzed the malware, naming the threat **Duqu**.[174][175] Symantec, based on this report, continued the analysis of the threat, calling it "nearly identical to Stuxnet, but with a completely different purpose", and published a detailed technical paper.[176] The main component used in Duqu is designed to capture information[70] such as keystrokes and system information. The exfiltrated data may be used to enable a future Stuxnet-like attack. On 28 December 2011, Kaspersky Lab's director of global research and analysis spoke to Reuters about recent research results showing that the platform Stuxnet and Duqu both originated in 2007, and is being referred to as Tilded due to the ~d at the beginning of the file names. Also uncovered in this research was the possibility for three more variants based on the Tilded platform.[177]

Flame

In May 2012, the new malware "Flame" was found, thought to be related to Stuxnet.[178] Researchers named the program "Flame" after the name of one of its modules.[178] After analysing the code of Flame, Kaspersky Lab said that there is a strong relationship between Flame and Stuxnet. An early version of Stuxnet contained code to propagate infections via USB drives that is nearly identical to a Flame module that exploits the same vulnerability.[179]

Media coverage

Since 2010, there has been extensive international news media coverage on Stuxnet and its aftermath. In early commentary, *The Economist* pointed out that Stuxnet was "a new kind of cyber-attack".[180] On 8 July 2011, *Wired* then published an article detailing how network security experts were able to decipher the origins of Stuxnet. In that piece, Kim Zetter claimed that Stuxnet's "cost–benefit ratio is still in question".[181] Later commentators tended to focus on the strategic significance of Stuxnet as a cyber weapon. Following the Wired piece, Holger Stark called Stuxnet the "first digital weapon of geopolitical importance, it could change the way wars are fought".[182] Meanwhile, Eddie Walsh referred to Stuxnet as "the world's newest high-end asymmetric threat".[183] Ultimately, some claim that the "extensive media coverage afforded to Stuxnet has only served as an advertisement for the vulnerabilities used by various cybercriminal groups".[184] While that may be the case, the media coverage has also increased awareness of cyber security threats.

Alex Gibney's 2016 documentary *Zero Days* covers the phenomenon around Stuxnet.[185] A zero-day (also known as 0-day) vulnerability is a computer-software vulnerability that is unknown to, or unaddressed by, those who should be interested in mitigating the vulnerability (including the vendor of the target software). Until the vulnerability is mitigated, hackers can exploit it to adversely affect computer programs, data, additional computers or a network.

In 2016, it was revealed that General James Cartwright, the former head of the U.S. Strategic Command, had leaked information related to Stuxnet. He later pleaded guilty for lying to FBI agents pursuing an investigation into the leak.[186][187] On 17 January 2017, he was granted a full pardon in this case by President Obama, thus expunging his conviction.

Darknet Diaries' Podcast episode Stuxnet (https://darknetdiaries.com/episode/29/), discusses Stuxnet with guest Kim Zetter and references the book Count Down to Zero Day.[188]

In popular culture

Besides the aforementioned Alex Gibney documentary *Zero Days* (2016), which looks into the malware and the cyberwarfare surrounding it, other works which reference Stuxnet include:

- In *Castle*, season 8, episode 18 "Backstabber" Stuxnet is revealed to have been (fictionally) created by MI6, and a version of it is used to take down the London power grid.
- *Trojan Horse* is a novel written by Windows utility writer and novelist Mark Russinovich. It features the usage of the Stuxnet virus as a main plot line for the story, and the attempt of Iran to bypass it.
- In *Ghost in the Shell: Arise*, Stuxnet is the named type of computer virus which infected Kusanagi and Manamura allowing false memories to be implanted.
- In July 2017, MRSA (Mat Zo) released a track named "Stuxnet" through Hospital Records.
- In Ubisoft's 2013 video game *Tom Clancy's Splinter Cell: Blacklist*, the protagonist, Sam Fisher, makes use of a mobile, airborne headquarters ("Paladin") which is said at one point within the game's story mode to have been targeted by a Stuxnet-style virus, causing its systems to fail and the plane to careen towards the ocean, and would have crashed without Fisher's intervening.[189]
- In Michael Mann's 2015 movie *Blackhat*, the code shown as belonging to a virus used by a hacker to cause the coolant pumps explosion in a nuclear plant in Chai Wan, Hong Kong, is actual Stuxnet decompiled code.
- In the third episode of *Star Trek: Discovery*, "Context Is for Kings", characters identify a segment of code as being part of an experimental transportation system. The code shown is decompiled Stuxnet code.[190] Much of the same code is shown in the episode "Pyre" of *The Expanse*, this time as a visual representation of a "diagnostic exploit" breaking into the control software for nuclear missiles.

See also

- 2024 Lebanon pager explosions
- Advanced persistent threat
- DigiNotar
- Killer poke
- List of security hacking incidents
- Mahdi (malware)
- Natanz
- Nitro Zeus
- Operation High Roller
- Operation Merlin
- Pin control attack
- Programmable logic controller
- Regin (malware)
- Stars virus
- Tailored Access Operations

- Vulnerability of nuclear plants to attack
- Zero Days

References

1. "W32.Stuxnet Dossier" (https://web.archive.org/web/20191104195500/https://www.wired.com/images_blogs/threatlevel/2010/11/w32_stuxnet_dossier.pdf) (PDF). Symantec. November 2010. Archived from the original (https://www.wired.com/images_blogs/threatlevel/2010/11/w32_stuxnet_dossier.pdf) (PDF) on 4 November 2019.
2. "Stuxnet : A worm which targets SCADA systems" (https://www.cert-ist.com/public/en/SO_detail?code=stuxnet). *CERT-IST Computer Emergency Response Team*. 8 September 2010. Retrieved 7 June 2025. "Stuxnet was discovered on June 17, 2010 by the Belarusian Company VirusBlokAda (a company that develops antivirus products). At that time most of the attention of the analysts was caught by the fact that this worm uses a previously unknown vulnerability in Windows (a "0-day" flaw): the ". LNK" vulnerability which led Microsoft to release early in August the out-of-band patch MS10-046. This is only after further analysis that analysts found that Stuxnet was in fact designed to target SCADA systems."
3. Kushner, David (26 February 2013). "The Real Story of Stuxnet". *IEEE Spectrum*. **50** (3): 48–53. Bibcode:2013IEEES..50c..48K (https://ui.adsabs.harvard.edu/abs/2013IEEES..50c..48K). doi:10.1109/MSPEC.2013.6471059 (https://doi.org/10.1109%2FMSPEC.2013.6471059). S2CID 29782870 (https://api.semanticscholar.org/CorpusID:29782870).
4. Sen, Ashish (10 April 2015). "Iran's Growing Cyber Capabilities in a Post-Stuxnet Era" (https://www.atlanticcouncil.org/blogs/new-atlanticist/iran-s-growing-cyber-capabilities-in-a-post-stuxnet-era/). *Atlantic Council*. Retrieved 3 September 2025.
5. "Confirmed: US and Israel created Stuxnet, lost control of it" (https://arstechnica.com/tech-policy/2012/06/confirmed-us-israel-created-stuxnet-lost-control-of-it/). *Ars Technica*. June 2012. Archived (https://web.archive.org/web/20190506093359/https://arstechnica.com/tech-policy/2012/06/confirmed-us-israel-created-stuxnet-lost-control-of-it/) from the original on 6 May 2019. Retrieved 15 June 2017.
6. Ellen Nakashima (2 June 2012). "Stuxnet was work of U.S. and Israeli experts, officials say" (https://www.washingtonpost.com/world/national-security/stuxnet-was-work-of-us-and-israeli-experts-officials-say/2012/06/01/gJQAlnEy6U_story.html). *The Washington Post*. Archived (https://web.archive.org/web/20190504083116/https://www.washingtonpost.com/world/national-security/stuxnet-was-work-of-us-and-israeli-experts-officials-say/2012/06/01/gJQAlnEy6U_story.html) from the original on 4 May 2019. Retrieved 8 September 2015.
7. Bergman, Ronen; Mazzetti, Mark (4 September 2019). "The Secret History of the Push to Strike Iran" (https://www.nytimes.com/2019/09/04/magazine/iran-strike-israel-america.html). *The New York Times Magazine*. ProQuest 2283858753 (https://www.proquest.com/docview/2283858753). Archived (https://web.archive.org/web/20230315042409/https://www.nytimes.com/2019/09/04/magazine/iran-strike-israel-america.html) from the original on 15 March 2023. Retrieved 23 March 2023.
8. Sanger, David E. (1 June 2012). "Obama Order Sped Up Wave of Cyberattacks Against Iran" (https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html). *The New York Times*. ISSN 0362-4331 (https://search.worldcat.org/issn/0362-4331). Archived (https://web.archive.org/web/20120601112345/http://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html) from the original on 1 June 2012. Retrieved 3 October 2022.

9. Naraine, Ryan (14 September 2010). "Stuxnet attackers used 4 Windows zero-day exploits" (https://web.archive.org/web/20141125225130/http://www.zdnet.com/blog/security/stuxnet-attackers-used-4-windows-zero-day-exploits/7347). *ZDNet*. Archived from the original (https://www.zdnet.com/blog/security/stuxnet-attackers-used-4-windows-zero-day-exploits/7347) on 25 November 2014. Retrieved 12 April 2014.
10. Karnouskos, Stamatis (November 2011). "Stuxnet worm impact on industrial cyber-physical system security" (https://papers.duckdns.org/files/2011_IECON_stuxnet.pdf) (PDF). *IECON 2011 – 37th Annual Conference of the IEEE Industrial Electronics Society*. pp. 4490–4494. doi:10.1109/IECON.2011.6120048 (https://doi.org/10.1109%2FIECON.2011.6120048). ISBN 978-1-61284-972-0. S2CID 1980890 (https://api.semanticscholar.org/CorpusID:1980890). Archived (https://web.archive.org/web/20230424052718/https://papers.duckdns.org/files/2011_IECON_stuxnet.pdf) (PDF) from the original on 24 April 2023. Retrieved 23 March 2023.
11. Kelley, Michael (20 November 2013). "The Stuxnet Attack on Iran's Nuclear Plant Was 'Far More Dangerous' Than Previously Thought" (http://www.businessinsider.com/stuxnet-was-far-more-dangerous-than-previous-thought-2013-11). *Business Insider*. Archived (https://web.archive.org/web/20140509020404/http://www.businessinsider.com/stuxnet-was-far-more-dangerous-than-previous-thought-2013-11) from the original on 9 May 2014. Retrieved 8 February 2014.
12. "Sheep dip your removable storage devices to reduce the threat of cyber attacks" (https://web.archive.org/web/20170904015230/https://www.mac-solutions.net/en/news/129-sheep-dip-your-removable-storage-devices-to-reduce-the-threat-of-cyber-attacks). *www.mac-solutions.net*. Archived from the original (https://www.mac-solutions.net/en/news/129-sheep-dip-your-removable-storage-devices-to-reduce-the-threat-of-cyber-attacks) on 4 September 2017. Retrieved 26 July 2017.
13. "STUXNET Malware Targets SCADA Systems" (http://about-threats.trendmicro.com/us/webattack/54/STUXNET%20Malware%20Targets%20SCADA%20Systems). Trend Micro. January 2012. Archived (https://web.archive.org/web/20140413144254/http://about-threats.trendmicro.com/us/webattack/54/STUXNET%20Malware%20Targets%20SCADA%20Systems) from the original on 13 April 2014. Retrieved 12 April 2014.
14. Gross, Michael Joseph (April 2011). "A Declaration of Cyber-War" (https://www.vanityfair.com/news/2011/03/stuxnet-201104). *Vanity Fair*. Archived (https://web.archive.org/web/20210831181802/https://www.vanityfair.com/news/2011/03/stuxnet-201104) from the original on 31 August 2021. Retrieved 31 December 2015.
15. "Exploring Stuxnet's PLC Infection Process" (https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=ad4b3d10-b808-414c-b4c3-ae4a2ed85560&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments). Symantec. 23 January 2014. Archived (https://web.archive.org/web/20210621221438/https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=ad4b3d10-b808-414c-b4c3-ae4a2ed85560&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments) from the original on 21 June 2021. Retrieved 22 September 2010.
16. "Building a Cyber Secure Plant" (https://web.archive.org/web/20210421092726/https://www.totallyintegratedautomation.com/2010/09/building-a-cyber-secure-plant/). *Totally Integrated Automation*. Siemens. 30 September 2010. Archived from the original (http://www.totallyintegratedautomation.com/2010/09/building-a-cyber-secure-plant/) on 21 April 2021. Retrieved 5 December 2010.
17. McMillan, Robert (16 September 2010). "Siemens: Stuxnet worm hit industrial systems" (http://www.computerworld.com/article/2515570/network-security/siemens--stuxnet-worm-hit-industrial-systems.html). *Computerworld*. IDG News. Archived (https://web.archive.org/web/20190220034909/https://www.computerworld.com/article/2515570/network-security/siemens--stuxnet-worm-hit-industrial-systems.html) from the original on 20 February 2019. Retrieved 16 September 2010.

18. "Last-minute paper: An indepth look into Stuxnet" (https://web.archive.org/web/20211209121553/https://www.virusbulletin.com/conference/vb2010/abstracts/indepth-look-stuxnet). Virus Bulletin. Archived from the original (https://www.virusbulletin.com/conference/vb2010/abstracts/indepth-look-stuxnet) on 9 December 2021.
19. "Stuxnet worm hits Iran nuclear plant staff computers" (https://web.archive.org/web/20170716012344/http://www.bbc.co.uk/news/world-middle-east-11414483). *BBC News*. 26 September 2010. Archived from the original (https://www.bbc.co.uk/news/world-middle-east-11414483) on 16 July 2017.
20. Nicolas Falliere (6 August 2010). "Stuxnet Introduces the First Known Rootkit for Industrial Control Systems" (http://www.symantec.com/connect/blogs/stuxnet-introduces-first-known-rootkit-scada-devices). Symantec. Retrieved 9 February 2011.
21. "Iran's Nuclear Agency Trying to Stop Computer Worm" (https://www.nytimes.com/aponline/2010/09/25/world/middleeast/AP-ML-Iran-Cyber-Attacks.html). Tehran. Associated Press. 25 September 2010. Retrieved 25 September 2010.
22. Keizer, Gregg (16 September 2010). "Is Stuxnet the 'best' malware ever?" (http://www.infoworld.com/article/2626009/malware/is-stuxnet-the--best--malware-ever-.html). *InfoWorld*. Archived (https://web.archive.org/web/20210505170423/https://www.infoworld.com/article/2626009/is-stuxnet-the--best--malware-ever-.html) from the original on 5 May 2021. Retrieved 16 September 2010.
23. Cherry, Steven; Langner, Ralph (13 October 2010). "How Stuxnet Is Rewriting the Cyberterrorism Playbook" (https://web.archive.org/web/20210414155345/https://spectrum.ieee.org/podcast/telecom/security/how-stuxnet-is-rewriting-the-cyberterrorism-playbook). *IEEE Spectrum*. Archived from the original (https://spectrum.ieee.org/podcast/telecom/security/how-stuxnet-is-rewriting-the-cyberterrorism-playbook) on 14 April 2021. Retrieved 2 February 2020.
24. "Stuxnet Virus Targets and Spread Revealed" (https://www.bbc.co.uk/news/technology-12465688). *BBC News*. 15 February 2011. Archived (https://web.archive.org/web/20211125040044/https://www.bbc.co.uk/news/technology-12465688) from the original on 25 November 2021. Retrieved 17 February 2011.
25. Fildes, Jonathan (23 September 2010). "Stuxnet worm 'targeted high-value Iranian assets'" (https://www.bbc.co.uk/news/technology-11388018). *BBC News*. Archived (https://web.archive.org/web/20100924033904/http://www.bbc.co.uk/news/technology-11388018) from the original on 24 September 2010. Retrieved 23 September 2010.
26. Beaumont, Claudine (23 September 2010). "Stuxnet virus: worm 'could be aimed at high-profile Iranian targets'" (https://www.telegraph.co.uk/technology/news/8021102/Stuxnet-virus-worm-could-be-aimed-at-high-profile-Iranian-targets.html). *The Daily Telegraph*. London. Archived (https://ghostarchive.org/archive/20220112/https://www.telegraph.co.uk/technology/news/8021102/Stuxnet-virus-worm-could-be-aimed-at-high-profile-Iranian-targets.html) from the original on 12 January 2022. Retrieved 28 September 2010.
27. MacLean, William (24 September 2010). "Update 2-Cyber attack appears to target Iran-tech firms" (https://www.reuters.com/article/idUSLDE68N1OI20100924). *Reuters*. Archived (https://web.archive.org/web/20211114140035/https://www.reuters.com/article/idUSLDE68N1OI20100924) from the original on 14 November 2021. Retrieved 2 July 2017.
28. "Iran Confirms Stuxnet Worm Halted Centrifuges" (https://www.cbsnews.com/news/iran-confirms-stuxnet-worm-halted-centrifuges/). *CBS News*. 29 November 2010. Archived (https://web.archive.org/web/20220512080538/https://www.cbsnews.com/news/iran-confirms-stuxnet-worm-halted-centrifuges/) from the original on 12 May 2022. Retrieved 12 May 2022.
29. Bronner, Ethan; Broad, William J. (29 September 2010). "In a Computer Worm, a Possible Biblical Clue" (https://www.nytimes.com/2010/09/30/world/middleeast/30worm.html?_r=2&pagewanted=2&hpw). *The New York Times*. Archived (https://web.archive.org/web/20220925044718/http://www.nytimes.com/2010/09/30/world/middleeast/30worm.html?_r=2&pagewanted=2&hpw) from the original on 25 September 2022. Retrieved 2 October 2010.

30. "Software smart bomb fired at Iranian nuclear plant: Experts" (http://economictimes.indiatimes.com/news/news-by-industry/et-cetera/Software-smart-bomb-fired-at-Iranian-nuclear-plant-Experts/articleshow/6617686.cms). Economictimes.indiatimes.com. 24 September 2010. Archived (https://web.archive.org/web/20211114141129/https://economictimes.indiatimes.com/news/news-by-industry/et-cetera/Software-smart-bomb-fired-at-Iranian-nuclear-plant-Experts/articleshow/6617686.cms) from the original on 14 November 2021. Retrieved 28 September 2010.
31. "Kaspersky Lab provides its insights on Stuxnet worm" (http://www.kaspersky.com/about/news/virus/2010/Kaspersky_Lab_provides_its_insights_on_Stuxnet_worm). *Kaspersky*. Russia. 24 September 2010. Archived (https://web.archive.org/web/20211116092936/https://www.kaspersky.com/about/press-releases) from the original on 16 November 2021. Retrieved 7 November 2011.
32. "Stuxnet Questions and Answers – F-Secure Weblog" (https://web.archive.org/web/20210505195313/https://archive.f-secure.com/weblog/archives/00002040.html). *F-Secure*. Finland. 1 October 2010. Archived from the original (http://www.f-secure.com/weblog/archives/00002040.html) on 5 May 2021.
33. Gary Samore (https://www.pbs.org/wnet/need-to-know/security/video-cracking-the-code-defending-against-the-superweapons-of-the-21st-century-cyberwar/9456/) Archived (https://web.archive.org/web/20180427015157/http://www.pbs.org/wnet/need-to-know/security/video-cracking-the-code-defending-against-the-superweapons-of-the-21st-century-cyberwar/9456/) 27 April 2018 at the Wayback Machine speaking at the 10 December 2010 Washington Forum of the Foundation for Defense of Democracies in Washington DC, reported by C-Span and contained in the PBS program *Need to Know* ("Cracking the code: Defending against the superweapons of the 21st century cyberwar" (https://web.archive.org/web/20110522023118/http://www.pbs.org/wnet/need-to-know/security/video-cracking-the-code-defending-against-the-superweapons-of-the-21st-century-cyberwar/9456/), 4 minutes into piece)
34. Williams, Christopher (15 February 2011). "Israel video shows Stuxnet as one of its successes" (https://www.telegraph.co.uk/news/worldnews/middleeast/israel/8326387/Israel-video-shows-Stuxnet-as-one-of-its-successes.html). London: Telegraph.co.uk. Archived (https://ghostarchive.org/archive/20220112/https://www.telegraph.co.uk/news/worldnews/middleeast/israel/8326387/Israel-video-shows-Stuxnet-as-one-of-its-successes.html) from the original on 12 January 2022. Retrieved 14 February 2012.
35. Sanger, David E. (1 June 2012). "Obama Order Sped Up Wave of Cyberattacks Against Iran" (https://web.archive.org/web/20170225103053/https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html). *The New York Times*. Archived from the original (https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html) on 25 February 2017. Retrieved 1 June 2012.
36. Matyszczyk, Chris (24 July 2012). "Thunderstruck! A tale of malware, AC/DC, and Iran's nukes" (https://news.cnet.com/8301-17852_3-57478515-71/thunderstruck-a-tale-of-malware-ac-dc-and-irans-nukes/). *CNET*. Retrieved 8 July 2013.
37. "Iran 'fends off new Stuxnet cyber attack'" (https://web.archive.org/web/20160807095642/https://www.bbc.com/news/world-middle-east-20842113). *BBC News*. 25 December 2012. Archived from the original (https://www.bbc.com/news/world-middle-east-20842113) on 7 August 2016. Retrieved 28 May 2015.
38. Shamah, David (11 November 2013). "Stuxnet, gone rogue, hit Russian nuke plant, space station" (https://web.archive.org/web/20170920120029/http://www.timesofisrael.com/stuxnet-gone-rogue-hit-russian-nuke-plant-space-station/). *The Times of Israel*. Archived from the original (https://www.timesofisrael.com/stuxnet-gone-rogue-hit-russian-nuke-plant-space-station/) on 20 September 2017. Retrieved 12 November 2013.

39. Krebs, Brian (17 July 2010). "Experts Warn of New Windows Shortcut Flaw" (http://krebsonsecurity.com/2010/07/experts-warn-of-new-windows-shortcut-flaw/). *Krebs on Security*. Archived (https://web.archive.org/web/20220902123724/https://krebsonsecurity.com/2010/07/experts-warn-of-new-windows-shortcut-flaw/) from the original on 2 September 2022. Retrieved 3 March 2011.
40. Gross, Michael Joseph (April 2011). "A Declaration of Cyber-War" (https://www.vanityfair.com/news/2011/03/stuxnet-201104). *Vanity Fair*. Condé Nast. Archived (https://web.archive.org/web/20210831181802/https://www.vanityfair.com/news/2011/03/stuxnet-201104) from the original on 31 August 2021. Retrieved 31 December 2015.
41. "Rootkit.TmpHider" (http://www.wilderssecurity.com/showthread.php?p=1712134#post1712134). *wilderssecurity.com*. Wilders Security Forums. Archived (https://web.archive.org/web/20131215102221/http://www.wilderssecurity.com/showthread.php?p=1712134#post1712134) from the original on 15 December 2013. Retrieved 25 March 2014.
42. Shearer, Jarrad (13 July 2010). "W32.Stuxnet" (https://web.archive.org/web/20120104215049/http://www.symantec.com/security_response/writeup.jsp?docid=2010-071400-3123-99). *Symantec*. Symantec. Archived from the original (http://www.symantec.com/security_response/writeup.jsp?docid=2010-071400-3123-99) on 4 January 2012. Retrieved 25 March 2014.
43. Zetter, Kim (11 July 2011). "How digital detectives deciphered Stuxnet, the most menacing malware in history" (https://arstechnica.com/tech-policy/2011/07/how-digital-detectives-deciphered-stuxnet-the-most-menacing-malware-in-history/). *arstechnica.com*. Archived (https://web.archive.org/web/20220514060226/https://arstechnica.com/tech-policy/2011/07/how-digital-detectives-deciphered-stuxnet-the-most-menacing-malware-in-history/) from the original on 14 May 2022. Retrieved 25 March 2014.
44. Karl (26 October 2011). "Stuxnet opens cracks in Iran nuclear program" (http://www.abc.net.au/science/articles/2011/10/26/3348123.htm). *abc.net.au*. ABC. Archived (https://web.archive.org/web/20210224154457/http://www.abc.net.au/science/articles/2011/10/26/3348123.htm) from the original on 24 February 2021. Retrieved 25 March 2014.
45. Gostev, Alexander (26 September 2010). "Myrtus and Guava: the epidemic, the trends, the numbers" (https://web.archive.org/web/20110101113112/http://www.securelist.com/en/blog/325/Myrtus_and_Guava_the_epidemic_the_trends_the_numbers). Archived from the original (http://www.securelist.com/en/blog/325/Myrtus_and_Guava_the_epidemic_the_trends_the_numbers) on 1 January 2011. Retrieved 22 January 2011.
46. Finkle, Jim (26 February 2013). "Researchers say Stuxnet was deployed against Iran in 2007" (https://www.reuters.com/article/us-cyberwar-stuxnet-idUSBRE91P0PP20130226). *Reuters*. Archived (https://web.archive.org/web/20210815220007/https://www.reuters.com/article/us-cyberwar-stuxnet-idUSBRE91P0PP20130226) from the original on 15 August 2021. Retrieved 6 July 2021.
47. Aleksandr Matrosov; Eugene Rodionov; David Harley & Juraj Malcho. "Stuxnet Under the Microscope, Revision 1.31" (https://web.archive.org/web/20220122123326/https://www.welivesecurity.com/wp-content/uploads/2012/11/Stuxnet_Under_the_Microscope.pdf) (PDF). Archived from the original (https://www.welivesecurity.com/media_files/white-papers/Stuxnet_Under_the_Microscope.pdf) (PDF) on 22 January 2022. Retrieved 6 September 2019.
48. Kiley, Sam (25 November 2010). "Super Virus A Target For Cyber Terrorists" (http://news.sky.com/skynews/Home/World-News/Stuxnet-Worm-Virus-Targeted-At-Irans-Nuclear-Plant-Is-In-Hands-Of-Bad-Guys-Sky-News-Sources-Say/Article/201011415827544). Archived (https://web.archive.org/web/20101128093507/http://news.sky.com/skynews/Home/World-News/Stuxnet-Worm-Virus-Targeted-At-Irans-Nuclear-Plant-Is-In-Hands-Of-Bad-Guys-Sky-News-Sources-Say/Article/201011415827544) from the original on 28 November 2010. Retrieved 25 November 2010.
49. "A Fanny Equation: 'I am your father, Stuxnet'" (https://securelist.com/a-fanny-equation-i-am-your-father-stuxnet/68787/). *Kaspersky Lab*. 17 February 2015. Archived (https://web.archive.org/web/20210319221224/https://securelist.com/a-fanny-equation-i-am-your-father-stuxnet/68787/) from the original on 19 March 2021. Retrieved 24 November 2015.

50. "fanny.bmp code" (https://github.com/loneicewolf/fanny.bmp/). *GitHub*. 23 October 2021. Archived (https://web.archive.org/web/20210203123746/https://github.com/loneicewolf/fanny.bmp) from the original on 3 February 2021. Retrieved 15 February 2021.
51. "Equation Group Questions and Answers" (https://web.archive.org/web/20150217023145/https://securelist.com/files/2015/02/Equation_group_questions_and_answers.pdf) (PDF). *securelist.com*. Archived from the original (https://securelist.com/files/2015/02/Equation_group_questions_and_answers.pdf) (PDF) on 17 February 2015.
52. Seals, Tara (9 April 2019). "SAS 2019: Stuxnet-Related APTs Form Gossip Girl, an 'Apex Threat Actor'" (https://threatpost.com/stuxnet-apts-gossip-girl/143595/). *threatpost.com*. Archived (https://web.archive.org/web/20200728022133/https://threatpost.com/stuxnet-apts-gossip-girl/143595/) from the original on 28 July 2020. Retrieved 6 August 2020.
53. Chronicle (12 April 2019). "Who is GOSSIPGIRL?" (https://medium.com/chronicle-blog/who-is-gossipgirl-3b4170f846c0). *Medium*. Archived (https://web.archive.org/web/20200722082746/https://medium.com/chronicle-blog/who-is-gossipgirl-3b4170f846c0) from the original on 22 July 2020. Retrieved 15 July 2020.
54. "CSEC SIGINT Cyber Discovery: Summary of the current effort" (https://web.archive.org/web/20150323101629/https://s3.amazonaws.com/s3.documentcloud.org/documents/1690221/doc-4-csec-sigint-discocon-2010.pdf) (PDF). *Electrospaces*. November 2010. Archived from the original (https://s3.amazonaws.com/s3.documentcloud.org/documents/1690221/doc-4-csec-sigint-discocon-2010.pdf) (PDF) on 23 March 2015.
55. Bencsáth, Boldizsár. "Territorial Dispute – NSA's perspective on APT landscape" (https://web.archive.org/web/20220110210924/https://www.crysys.hu/publications/files/tedi/ukatemicrysys_territorialdispute.pdf) (PDF). Archived from the original (https://www.crysys.hu/publications/files/tedi/ukatemicrysys_territorialdispute.pdf) (PDF) on 10 January 2022.
56. Marschalek, Marion; Guarnieri, Claudio (25 August 2015). "Big Game Hunting: The Peculiarities of Nation-State Malware Research" (https://www.youtube.com/watch?v=hVlMYXLB0m0). *YouTube*. Archived (https://ghostarchive.org/varchive/youtube/20211221/hVlMYXLB0m0) from the original on 21 December 2021.
57. Barth, Bradley (10 April 2019). "GOSSIPGIRL – Stuxnet group had '4th man;' unknown version of Flame & Duqu found" (https://www.scmagazineuk.com/gossipgirl-stuxnet-group-4th-man-unknown-version-flame-duqu-found/article/1581579). Archived (https://web.archive.org/web/20200806024942/https://www.scmagazineuk.com/gossipgirl-stuxnet-group-4th-man-unknown-version-flame-duqu-found/article/1581579) from the original on 6 August 2020.
58. BetaFred. "Microsoft Security Bulletin MS10-061 – Critical" (https://docs.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-061). *docs.microsoft.com*. Archived (https://web.archive.org/web/20201006130903/https://docs.microsoft.com/en-us/security-updates/securitybulletins/2010/ms10-061) from the original on 6 October 2020. Retrieved 29 September 2020.
59. BetaFred. "Microsoft Security Bulletin MS08-067 – Critical" (https://docs.microsoft.com/en-us/security-updates/securitybulletins/2008/ms08-067). *docs.microsoft.com*. Archived (https://web.archive.org/web/20201206031538/https://docs.microsoft.com/en-us/security-updates/SecurityBulletins/2008/ms08-067) from the original on 6 December 2020. Retrieved 29 September 2020.
60. fmm (28 September 2020). "The Emerald Connection: EquationGroup collaboration with Stuxnet" (https://web.archive.org/web/20200930073726/https://fmmresearch.wordpress.com/2020/09/28/the-emerald-connection-equationgroup-collaboration-with-stuxnet/). *Facundo Muñoz Research*. Archived from the original (https://fmmresearch.wordpress.com/2020/09/28/the-emerald-connection-equationgroup-collaboration-with-stuxnet/) on 30 September 2020. Retrieved 29 September 2020.
61. "W32.Stuxnet" (https://web.archive.org/web/20120104215049/http://www.symantec.com/security_response/writeup.jsp?docid=2010-071400-3123-99). Symantec. 17 September 2010. Archived from the original (http://www.symantec.com/security_response/writeup.jsp?docid=2010-071400-3123-99) on 4 January 2012. Retrieved 2 March 2011.

62. "Iran denies hacking into American banks (https://www.reuters.com/article/us-iran-cyberattacks-denial-idUSBRE88M06O20120923) Archived (https://web.archive.org/web/20150924170905/http://www.reuters.com/article/2012/09/23/us-iran-cyberattacks-denial-idUSBRE88M06O20120923) 24 September 2015 at the Wayback Machine" Reuters, 23 September 2012
63. "Iranian Offensive Cyberattack Capabilities" (https://www.congress.gov/crs_external_products/IF/HTML/IF11406.web.html). *www.congress.gov*. 13 January 2020. Retrieved 21 September 2025.
64. "Operation Ababil (2012) | Research Starters | EBSCO Research" (https://www.ebsco.com/). *EBSCO*. Retrieved 3 September 2025.
65. "Compromise of Saudi Aramco and RasGas | CFR Interactives" (https://www.cfr.org/cyber-operations/compromise-saudi-aramco-and-rasgas). *www.cfr.org*. Retrieved 3 September 2025.
66. "Shamoon – Darknet Diaries" (https://darknetdiaries.com/transcript/30/). *darknetdiaries.com*. Retrieved 3 September 2025.
67. "Las Vegas Sands' network hit by destructive malware in Feb: Bloomberg" (https://www.reuters.com/article/technology/las-vegas-sands-network-hit-by-destructive-malware-in-feb-bloomberg-idUSKBN0JQ045/). *Reuters*. 12 December 2014. Retrieved 3 September 2025.
68. "Las Vegas Sands' Casino Network hit by Destructive Malware" (https://thehackernews.com/2014/12/las-vegas-casino-hacked.html). *The Hacker News*. Retrieved 3 September 2025.
69. Broad, William J.; Markoff, John; Sanger, David E. (15 January 2011). "Israel Tests on Worm Called Crucial in Iran Nuclear Delay" (https://www.nytimes.com/2011/01/16/world/middleeast/16stuxnet.html?_r=1&ref=general&src=me&pagewanted=all). *New York Times*. Archived (https://web.archive.org/web/20110920071931/http://www.nytimes.com/2011/01/16/world/middleeast/16stuxnet.html?_r=1&ref=general&src=me&pagewanted=all) from the original on 20 September 2011. Retrieved 16 January 2011.
70. Steven Cherry; with Larry Constantine (14 December 2011). "Sons of Stuxnet" (https://web.archive.org/web/20210414174152/https://spectrum.ieee.org/podcast/telecom/security/sons-of-stuxnet). *IEEE Spectrum*. Archived from the original (https://spectrum.ieee.org/podcast/telecom/security/sons-of-stuxnet) on 14 April 2021. Retrieved 2 February 2020.
71. "Conficker Worm: Help Protect Windows from Conficker" (https://technet.microsoft.com/en-us/security/dd452420.aspx). Microsoft. 10 April 2009. Archived (https://web.archive.org/web/20180518213908/https://technet.microsoft.com/en-us/security/dd452420.aspx) from the original on 18 May 2018. Retrieved 6 December 2010.
72. Buda, Alex (4 December 2016). "Creating Malware using the Stuxnet LNK Exploit" (http://www.rubydevices.com.au/blog/stuxnetLnkExploit). Ruby Devices. Archived (https://web.archive.org/web/20170318174047/http://www.rubydevices.com.au/blog/stuxnetLnkExploit) from the original on 18 March 2017. Retrieved 18 March 2017.
73. Kim Zetter (23 September 2010). "Blockbuster Worm Aimed for Infrastructure, But No Proof Iran Nukes Were Target" (https://www.wired.com/2010/09/stuxnet-2/). *Wired*. Archived (https://web.archive.org/web/20161105160748/https://www.wired.com/2010/09/stuxnet-2/) from the original on 5 November 2016. Retrieved 4 November 2016.
74. Liam O Murchu (17 September 2010). "Stuxnet P2P component" (https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=12adb5c4-1b6b-41dc-95a5-e6320371a847&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments). Symantec. Archived (https://web.archive.org/web/20190117171829/https://www.symantec.com/connect/blogs/stuxnet-p2p-component) from the original on 17 January 2019. Retrieved 24 September 2010.
75. "W32.Stuxnet Dossier" (https://web.archive.org/web/20120707184843/http://www.symantec.com/content/en/us/enterprise/media/security_response/whitepapers/w32_stuxnet_dossier.pdf) (PDF). Symantec Corporation. Archived from the original (http://www.symantec.com/content/en/us/enterprise/media/security_response/whitepapers/w32_stuxnet_dossier.pdf) (PDF) on 7 July 2012. Retrieved 1 October 2010.

76. Microsoft (14 September 2010). "Microsoft Security Bulletin MS10-061 – Critical" (https://technet.microsoft.com/en-us/library/security/ms10-061.aspx). Microsoft. Archived (https://web.archive.org/web/20150320051551/https://technet.microsoft.com/en-us/library/security/ms10-061.aspx) from the original on 20 March 2015. Retrieved 20 August 2015.
77. Microsoft (2 August 2010). "Microsoft Security Bulletin MS10-046 – Critical" (https://technet.microsoft.com/en-us/library/security/ms10-046.aspx). Microsoft. Archived (https://web.archive.org/web/20150812084743/https://technet.microsoft.com/en-us/library/security/ms10-046.aspx) from the original on 12 August 2015. Retrieved 20 August 2015.
78. Gostev, Alexander (14 September 2010). "Myrtus and Guava, Episode MS10-061" (https://securelist.com/blog/incidents/29747/myrtus-and-guava-episode-ms10-061/). Kaspersky Lab. Archived (https://web.archive.org/web/20150823200012/https://securelist.com/blog/incidents/29747/myrtus-and-guava-episode-ms10-061/) from the original on 23 August 2015. Retrieved 20 August 2015.
79. "Kaspersky Lab provides its insights on Stuxnet worm" (http://www.kaspersky.com/about/news/virus/2010/Kaspersky_Lab_provides_its_insights_on_Stuxnet_worm). Kaspersky Lab. 24 September 2010. Archived (https://web.archive.org/web/20211116092936/https://www.kaspersky.com/about/press-releases) from the original on 16 November 2021. Retrieved 27 September 2010.
80. Gross, Michael Joseph (April 2011). "A Declaration of Cyber-War" (https://www.vanityfair.com/news/2011/03/stuxnet-201104). *Vanity Fair*. Archived (https://web.archive.org/web/20210831181802/https://www.vanityfair.com/news/2011/03/stuxnet-201104) from the original on 31 August 2021. Retrieved 4 March 2011.
81. Langner, Ralph (14 September 2010). "Ralph's Step-By-Step Guide to Get a Crack at Stuxnet Traffic and Behaviour" (http://www.langner.com/en/2010/09/14/ralphs-step-by-step-guide-to-get-a-crack-at-stuxnet-traffic-and-behavior/). *Ot-Base by Langner*. Archived (https://web.archive.org/web/20160625153256/http://www.langner.com/en/2010/09/14/ralphs-step-by-step-guide-to-get-a-crack-at-stuxnet-traffic-and-behavior/) from the original on 25 June 2016. Retrieved 4 March 2011.
82. Falliere, Nicolas (26 September 2010). "Stuxnet Infection of Step 7 Projects" (https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=0c21cbc7-0c40-44e7-8539-8d682b80103c&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments). Symantec. Archived (https://web.archive.org/web/20150103013845/http://www.symantec.com/connect/blogs/stuxnet-infection-step-7-projects) from the original on 3 January 2015. Retrieved 9 February 2011.
83. "Vulnerability Summary for CVE-2010-2772" (http://web.nvd.nist.gov/view/vuln/detail?vulnId=CVE-2010-2772). National Vulnerability Database. 22 July 2010. Archived (https://web.archive.org/web/20100811213718/http://web.nvd.nist.gov/view/vuln/detail?vulnId=CVE-2010-2772) from the original on 11 August 2010. Retrieved 7 December 2010.
84. Chien, Eric (12 November 2010). "Stuxnet: A Breakthrough" (https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=550505c5-c38a-4e0c-b590-f731bb3a60ad&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments). Symantec. Archived (https://web.archive.org/web/20180118011202/http://www.symantec.com/connect/blogs/stuxnet-breakthrough) from the original on 18 January 2018. Retrieved 14 November 2010.
85. "SIMATIC WinCC / SIMATIC PCS 7: Information concerning Malware / Virus / Trojan" (https://support.industry.siemens.com/cs/document/43876783/simatic-wincc-simatic-pcs-7%3A-information-about-malware-viruses-trojan-horses?dti=0&lc=en-WW). Siemens. Archived (https://web.archive.org/web/20190923212600/https://support.industry.siemens.com/cs/document/43876783/simatic-wincc-simatic-pcs-7:-information-about-malware-viruses-trojan-horses%3Fdti%3D0%26lc%3Den-WW) from the original on 23 September 2019. Retrieved 24 September 2010.

86. Espiner, Tom (20 July 2010). "Siemens warns Stuxnet targets of password risk" (https://www.cnet.com/news/privacy/siemens-warns-stuxnet-targets-of-password-risk/). *CNET*. Archived (https://web.archive.org/web/20110109160245/http://news.cnet.com/8301-1009_3-20011095-83.html) from the original on 9 January 2011. Retrieved 23 March 2023.
87. crve (17 September 2010). "Stuxnet also found at industrial plants in Germany" (http://www.h-online.com/security/news/item/Stuxnet-also-found-at-industrial-plants-in-Germany-1081469.html). The H. Archived (https://web.archive.org/web/20100921075008/http://www.h-online.com/security/news/item/Stuxnet-also-found-at-industrial-plants-in-Germany-1081469.html) from the original on 21 September 2010. Retrieved 18 September 2010.
88. "Repository of Industrial Security Incidents" (http://www.securityincidents.org/). Security Incidents Organization. Archived (https://web.archive.org/web/20110426194554/http://www.securityincidents.org/) from the original on 26 April 2011. Retrieved 14 October 2010.
89. "DHS National Cyber Security Division's CSSP" (http://www.us-cert.gov/control_systems/). DHS. Archived (https://web.archive.org/web/20101008004511/http://www.us-cert.gov/control_systems/) from the original on 8 October 2010. Retrieved 14 October 2010.
90. "ISA99, Industrial Automation and Control System Security" (https://web.archive.org/web/20110110194234/http://www.isa.org/MSTemplate.cfm?MicrositeID=988&CommitteeID=6821). International Society of Automation. Archived from the original (http://www.isa.org/MSTemplate.cfm?MicrositeID=988&CommitteeID=6821) on 10 January 2011. Retrieved 14 October 2010.
91. "Industrial communication networks – Network and system security – Part 2-1: Establishing an industrial automation and control system security program" (http://www.iec.ch/cgi-bin/procgi.pl/www/iecwww.p?wwwlang=E&wwwprog=pro-det.p&He=IEC&Pu=62443&Pa=2&Se=1&Am=&Fr=&TR=&Ed=1). International Electrotechnical Commission. Retrieved 14 October 2010.
92. "Chemical Sector Cyber Security Program" (https://web.archive.org/web/20101019140808/http://americanchemistry.com/s_chemitc/sec_callout.asp?CID=1733&DID=6429). ACC ChemITC. Archived from the original (http://www.americanchemistry.com/s_chemitc/sec_callout.asp?CID=1733&DID=6429) on 19 October 2010. Retrieved 14 October 2010.
93. "Pipeline SCADA Security Standard" (http://www.api.org/meetings/topics/pipeline/upload/Pipeline_SCADA_Security_Standard.pdf) (PDF). API. Archived (https://web.archive.org/web/20101119190725/http://api.org/meetings/topics/pipeline/upload/Pipeline_SCADA_Security_Standard.pdf) (PDF) from the original on 19 November 2010. Retrieved 19 November 2010.
94. Marty Edwards (Idaho National Laboratory) & Todd Stauffer (Siemens). *2008 Automation Summit: A User's Conference* (https://graphics8.nytimes.com/packages/pdf/science/NSTB.pdf) (PDF). United States Department of Homeland Security. p. 35. Archived (https://web.archive.org/web/20110120032529/http://graphics8.nytimes.com/packages/pdf/science/NSTB.pdf) (PDF) from the original on 20 January 2011. Retrieved 18 January 2011.
95. "The Can of Worms Is Open-Now What?" (http://www.controlglobal.com/articles/2010/StuxnetWorm1010.html). controlglobal.com. Archived (https://web.archive.org/web/20101001122724/http://www.controlglobal.com/articles/2010/StuxnetWorm1010.html) from the original on 1 October 2010. Retrieved 14 October 2010.
96. Byres, Eric; Cusimano, John (16 February 2012). "The 7 Steps to ICS Security" (https://web.archive.org/web/20130123141949/http://www.exida.com/index.php/News/new_whitepaper_the_7_steps_to_ics_and_scada_security/). Tofino Security and exida Consulting LLC. Archived from the original (http://www.exida.com/index.php/News/new_whitepaper_the_7_steps_to_ics_and_scada_security/) on 23 January 2013. Retrieved 3 March 2011.
97. Halliday, Josh (24 September 2010). "Stuxnet worm is the 'work of a national government agency'" (https://www.theguardian.com/technology/2010/sep/24/stuxnet-worm-national-agency). *The Guardian*. London. Archived (https://web.archive.org/web/20220822123431/https://www.theguardian.com/technology/2010/sep/24/stuxnet-worm-national-agency) from the original on 22 August 2022. Retrieved 27 September 2010.

98. Markoff, John (26 September 2010). "A Silent Attack, but Not a Subtle One" (https://www.nytimes.com/2010/09/27/technology/27virus.html). *The New York Times*. Archived (https://web.archive.org/web/20210206055520/https://www.nytimes.com/2010/09/27/technology/27virus.html) from the original on 6 February 2021. Retrieved 27 September 2010.
99. Schneier, Bruce (6 October 2010). "The Story Behind The Stuxnet Virus" (https://www.forbes.com/2010/10/06/iran-nuclear-computer-technology-security-stuxnet-worm.html). *Forbes*. Archived (https://web.archive.org/web/20170830063315/https://www.forbes.com/2010/10/06/iran-nuclear-computer-technology-security-stuxnet-worm.html) from the original on 30 August 2017. Retrieved 22 August 2017.
100. Schneier, Bruce (23 February 2012). "Another Piece of the Stuxnet Puzzle" (http://www.schneier.com/blog/archives/2012/02/another_piece_o.html). Schneier on Security. Archived (https://web.archive.org/web/20120226224354/http://www.schneier.com/blog/archives/2012/02/another_piece_o.html) from the original on 26 February 2012. Retrieved 4 March 2012.
101. Modderkolk, Huib (8 January 2024). "Sabotage in Iran: Een missie in duisternis" (https://www.volkskrant.nl/kijkverder/v/2024/sabotage-in-iran-een-missie-in-duisternis~v989743/) [Sabotage in Iran: A Mission in Darkness]. *De Volksrant* (in Dutch). Archived (https://web.archive.org/web/20240108225855/https://www.volkskrant.nl/kijkverder/v/2024/sabotage-in-iran-een-missie-in-duisternis~v989743/) from the original on 8 January 2024. Retrieved 26 June 2025.
102. Waterfield, Bruno (8 January 2024). "Dutch spies hid engineer's role in paralysing Iran nuclear project" (https://www.thetimes.com/world/article/dutch-spies-hid-engineers-role-in-stalling-iran-nuclear-project-dtqh3s2hq). *The Times and The Sunday Times*. Retrieved 19 July 2025.
103. Kovacs, Eduard (10 January 2024). "Dutch Engineer Used Water Pump to Get Billion-Dollar Stuxnet Malware Into Iranian Nuclear Facility: Report" (https://www.securityweek.com/dutch-engineer-used-water-pump-to-get-billion-dollar-stuxnet-malware-into-iranian-nuclear-facility-report/). *Security Week*. Archived (https://web.archive.org/web/20240515023816/https://www.securityweek.com/dutch-engineer-used-water-pump-to-get-billion-dollar-stuxnet-malware-into-iranian-nuclear-facility-report/) from the original on 15 May 2024.
104. Bright, Arthur (1 October 2010). "Clues Emerge About Genesis of Stuxnet Worm" (https://www.csmonitor.com/World/terrorism-security/2010/1001/Clues-emerge-about-genesis-of-Stuxnet-worm). *Christian Science Monitor*. Archived (https://web.archive.org/web/20110306114249/http://www.csmonitor.com/World/terrorism-security/2010/1001/Clues-emerge-about-genesis-of-Stuxnet-worm) from the original on 6 March 2011. Retrieved 4 March 2011.
105. Langner, Ralph (February 2011). *Ralph Langner: Cracking Stuxnet, a 21st-century cyber weapon* (https://web.archive.org/web/20140201101050/http://www.ted.com/talks/ralph_langner_cracking_stuxnet_a_21st_century_cyberweapon.html) (video). TED. Archived from the original (https://www.ted.com/talks/ralph_langner_cracking_stuxnet_a_21st_century_cyber_weapon) on 1 February 2014. Retrieved 4 January 2023.
106. McMillan, Robert (23 July 2010). "Iran was prime target of SCADA worm" (http://www.computerworld.com/article/2519584/network-security/iran-was-prime-target-of-scada-worm.html). *Computerworld*. Archived (https://web.archive.org/web/20140905183534/http://www.computerworld.com/article/2519584/network-security/iran-was-prime-target-of-scada-worm.html) from the original on 5 September 2014. Retrieved 17 September 2010.
107. Woodward, Paul (22 September 2010). "Iran confirms Stuxnet found at Bushehr nuclear power plant" (http://warincontext.org/2010/09/26/iran-confirms-stuxnet-found-at-bushehr-nuclear-power-plant/). Warincontext.org. Archived (https://web.archive.org/web/20190320070804/http://warincontext.org/2010/09/26/iran-confirms-stuxnet-found-at-bushehr-nuclear-power-plant/) from the original on 20 March 2019. Retrieved 28 September 2010.
108. "6 mysteries about Stuxnet" (https://web.archive.org/web/20140209181744/http://blog.foreignpolicy.com/posts/2010/09/27/6_mysteries_about_stuxnet). Blog.foreignpolicy.com. Archived from the original (http://blog.foreignpolicy.com/posts/2010/09/27/6_mysteries_about_stuxnet) on 9 February 2014. Retrieved 28 September 2010.

109. Clayton, Mark (21 September 2010). "Stuxnet malware is 'weapon' out to destroy ... Iran's Bushehr nuclear plant?" (https://www.csmonitor.com/USA/2010/0921/Stuxnet-malware-is-weapon-out-to-destroy-Iran-s-Bushehr-nuclear-plant). *Christian Science Monitor*. Archived (https://web.archive.org/web/20100924065250/http://www.csmonitor.com/USA/2010/0921/Stuxnet-malware-is-weapon-out-to-destroy-Iran-s-Bushehr-nuclear-plant) from the original on 24 September 2010. Retrieved 23 September 2010.
110. Melman, Yossi (28 September 2010). "Computer virus in Iran actually targeted larger nuclear facility" (https://www.haaretz.com/print-edition/news/computer-virus-in-iran-actually-targeted-larger-nuclear-facility-1.316052). *Haaretz*. Archived (https://web.archive.org/web/20110122133842/http://www.haaretz.com/print-edition/news/computer-virus-in-iran-actually-targeted-larger-nuclear-facility-1.316052) from the original on 22 January 2011. Retrieved 1 January 2011.
111. Melman, Yossi (24 November 2010). "Iran pauses uranium enrichment at Natanz nuclear plant" (https://www.haaretz.com/news/international/iran-pauses-uranium-enrichment-at-natanz-nuclear-plant-1.326276). Haaretz. Archived (https://web.archive.org/web/20101124013831/http://www.haaretz.com/news/international/iran-pauses-uranium-enrichment-at-natanz-nuclear-plant-1.326276) from the original on 24 November 2010. Retrieved 24 November 2010.
112. "The Stuxnet worm: A cyber-missile aimed at Iran?" (https://www.economist.com/blogs/babbage/2010/09/stuxnet_worm). *The Economist*. 24 September 2010. Archived (https://web.archive.org/web/20100927170936/http://www.economist.com/blogs/babbage/2010/09/stuxnet_worm) from the original on 27 September 2010. Retrieved 28 September 2010.
113. "Serious nuclear accident may lay behind Iranian nuke chief's mystery resignation" (https://web.archive.org/web/20101230121529/http://mirror.wikileaks.info/wiki/Serious_nuclear_accident_may_lay_behind_Iranian_nuke_chief%27s_mystery_resignation/). WikiLeaks. 16 July 2009. Archived from the original (http://mirror.wikileaks.info/wiki/Serious_nuclear_accident_may_lay_behind_Iranian_nuke_chief%27s_mystery_resignation) on 30 December 2010. Retrieved 1 January 2011.
114. IAEA Report on Iran (http://www.isisnucleariran.org/assets/pdf/ISIS_Analysis_IAEA_Report_16Nov2009.pdf) (PDF) (Report). Institute for Science and International Security. 16 November 2010. Archived (https://web.archive.org/web/20110311033912/http://www.isisnucleariran.org/assets/pdf/ISIS_Analysis_IAEA_Report_16Nov2009.pdf) (PDF) from the original on 11 March 2011. Retrieved 1 January 2011.
115. "Did Stuxnet Take Out 1,000 Centrifuges at the Natanz Enrichment Plant?" (http://isis-online.org/uploads/isis-reports/documents/stuxnet_FEP_22Dec2010.pdf) (PDF). Institute for Science and International Security. 22 December 2010. Archived (https://web.archive.org/web/20120910192149/http://isis-online.org/uploads/isis-reports/documents/stuxnet_FEP_22Dec2010.pdf) (PDF) from the original on 10 September 2012. Retrieved 27 December 2010.
116. Stöcker, Christian (26 December 2010). "Stuxnet-Virus könnte tausend Uran-Zentrifugen zerstört haben" (http://www.spiegel.de/netzwelt/netzpolitik/0,1518,736604,00.html). *Der Spiegel* (in German). Archived (https://web.archive.org/web/20101227155849/http://www.spiegel.de/netzwelt/netzpolitik/0,1518,736604,00.html) from the original on 27 December 2010. Retrieved 27 December 2010.
117. Stark, Holger (8 August 2011). "Mossad's Miracle Weapon: Stuxnet Virus Opens New Era of Cyber War" (http://www.spiegel.de/international/world/0,1518,778912,00.html). *Der Spiegel*. Archived (https://web.archive.org/web/20110815150818/http://www.spiegel.de/international/world/0,1518,778912,00.html) from the original on 15 August 2011. Retrieved 15 August 2011.
118. Warrick, Joby (15 February 2011). "Iran's Natanz nuclear facility recovered quickly from Stuxnet cyberattack" (https://web.archive.org/web/20220124000904/https://www.washingtonpost.com/world/irans-natanz-nuclear-facility-recovered-quickly-from-stuxnet-cyber-attack/2011/02/15/ABUIkoQ_story.html). *The Washington Post*. Archived from the original (https://www.washingtonpost.com/world/irans-natanz-nuclear-facility-recovered-quickly-from-stuxnet-cyber-attack/2011/02/15/ABUIkoQ_story.html) on 24 January 2022. Retrieved 23 March 2023.

119. "Stuxnet Malware and Natanz: Update of ISIS December 22, 2010 Report" (http://isis-online.org/isis-reports/detail/stuxnet-malware-and-natanz-update-of-isis-december-22-2010-reportsupa-href1/8). Institute for Science and International Security. 15 February 2011. Archived (https://web.archive.org/web/20110807063612/http://isis-online.org/isis-reports/detail/stuxnet-malware-and-natanz-update-of-isis-december-22-2010-reportsupa-href1/8) from the original on 7 August 2011. Retrieved 10 July 2011.
120. "Signs of sabotage in Tehran's nuclear programme" (https://web.archive.org/web/20101120230019/http://gulfnews.com/news/region/iran/signs-of-sabotage-in-tehran-s-nuclear-programme-1.658481). *Gulf News*. 14 July 2010. Archived from the original (http://gulfnews.com/news/region/iran/signs-of-sabotage-in-tehran-s-nuclear-programme-1.658481) on 20 November 2010.
121. Williams, Dan (7 July 2009). "Wary of naked force, Israel eyes cyberwar on Iran" (https://www.reuters.com/article/idUSLV83872). *Reuters*. Archived (https://web.archive.org/web/20180519121118/https://www.reuters.com/article/idUSLV83872) from the original on 19 May 2018. Retrieved 2 July 2017.
122. Aneja, Atul (26 September 2010). "Under cyber-attack, says Iran" (http://www.thehindu.com/news/international/article797363.ece). *The Hindu*. Chennai, India. Archived (https://web.archive.org/web/20100929062917/http://www.thehindu.com/news/international/article797363.ece) from the original on 29 September 2010. Retrieved 27 September 2010.
123. "شبکه خبر :: راه های مقابله با ویروس"استاکس نت (https://web.archive.org/web/20130621130831/http://www.irinn.ir/Default.aspx?TabId=56&nid=193428) (in Persian). Irinn.ir. Archived from the original (http://www.irinn.ir/Default.aspx?TabId=56&nid=193428) on 21 June 2013. Retrieved 28 September 2010.
124. "Stuxnet worm rampaging through Iran: IT official" (https://web.archive.org/web/20100930235902/http://www.google.com/hostednews/afp/article/ALeqM5jnTuOwD-HMJB9A8JQUC_-E_o2IYw). AFP. Archived from the original (https://www.google.com/hostednews/afp/article/ALeqM5jnTuOwD-HMJB9A8JQUC_-E_o2IYw) on 30 September 2010.
125. "IRAN: Speculation on Israeli involvement in malware computer attack" (http://latimesblogs.latimes.com/babylonbeyond/2010/09/israel-questions-about-the-stuxnet-attack-on-iranian-computers.html). *Los Angeles Times*. 27 September 2010. Archived (https://web.archive.org/web/20100928045055/http://latimesblogs.latimes.com/babylonbeyond/2010/09/israel-questions-about-the-stuxnet-attack-on-iranian-computers.html) from the original on 28 September 2010. Retrieved 28 September 2010.
126. Erdbrink, Thomas; Nakashima, Ellen (27 September 2010). "Iran struggling to contain 'foreign-made' 'Stuxnet' computer virus" (https://www.washingtonpost.com/wp-dyn/content/article/2010/09/27/AR2010092706229.html). *The Washington Post*. Archived (https://web.archive.org/web/20101002041632/http://www.washingtonpost.com/wp-dyn/content/article/2010/09/27/AR2010092706229.html) from the original on 2 October 2010. Retrieved 28 September 2010.
127. "Ahmadinedschad räumt Virus-Attack ein" (http://www.spiegel.de/netzwelt/gadgets/0,1518,731881,00.html). *Der Spiegel*. 29 November 2010. Archived (https://web.archive.org/web/20101220052129/http://www.spiegel.de/netzwelt/gadgets/0,1518,731881,00.html) from the original on 20 December 2010. Retrieved 29 December 2010.
128. "Stuxnet: Ahmadinejad admits cyberweapon hit Iran nuclear program" (https://www.csmonitor.com/USA/2010/1130/Stuxnet-Ahmadinejad-admits-cyberweapon-hit-Iran-nuclear-program). *The Christian Science Monitor*. 30 November 2010. Archived (https://web.archive.org/web/20101205094238/http://www.csmonitor.com/USA/2010/1130/Stuxnet-Ahmadinejad-admits-cyberweapon-hit-Iran-nuclear-program) from the original on 5 December 2010. Retrieved 29 December 2010.

129. Zetter, Kim (29 November 2010). "Iran: Computer Malware Sabotaged Uranium Centrifuges | Threat Level" (https://www.wired.com/threatlevel/2010/11/stuxnet-sabotage-centrifuges/). *Wired*. Archived (https://web.archive.org/web/20120311084917/http://www.wired.com/threatlevel/2010/11/stuxnet-sabotage-centrifuges/) from the original on 11 March 2012. Retrieved 14 February 2012.
130. "US Denies Role in Iranian Scientist's Death" (https://www.foxnews.com/world/us-denies-role-in-iranian-scientists-death/). Fox News. 7 April 2010. Archived (https://web.archive.org/web/20120213004215/http://www.foxnews.com/world/2012/01/11/report-bomb-kills-iran-university-professor/?test=latestnews) from the original on 13 February 2012. Retrieved 14 February 2012.
131. Monica Amarelo (21 January 2011). "New FAS Report Demonstrates Iran Improved Enrichment in 2010" (https://web.archive.org/web/20131215104145/http://www.fas.org/press/news/2011/issuebrief_iran.html). Federation of American Scientists. Archived from the original (https://fas.org/press/news/2011/issuebrief_iran.html) on 15 December 2013. Retrieved 1 January 2016.
132. "Report: Iran's nuclear capacity unharmed, contrary to U.S. assessment" (https://www.haaretz.com/news/international/report-iran-s-nuclear-capacity-unharmed-contrary-to-u-s-assessment-1.338522). *Haaretz*. 22 January 2011. Archived (https://web.archive.org/web/20110125182209/http://www.haaretz.com/news/international/report-iran-s-nuclear-capacity-unharmed-contrary-to-u-s-assessment-1.338522) from the original on 25 January 2011. Retrieved 27 January 2011.
133. Jeffrey Goldberg (22 January 2011). "Report: Report: Iran's Nuclear Program Going Full Speed Ahead" (https://www.theatlantic.com/international/archive/2011/01/report-irans-nuclear-program-going-full-speed-ahead/70031/). *The Atlantic*. Archived (https://web.archive.org/web/20161112025144/http://www.theatlantic.com/international/archive/2011/01/report-irans-nuclear-program-going-full-speed-ahead/70031/) from the original on 12 November 2016. Retrieved 11 March 2017.
134. "Experts say Iran has 'neutralized' Stuxnet virus" (https://www.reuters.com/article/us-iran-usa-stuxnet-idUSTRE81D24Q20120214). *Reuters*. 14 February 2012. Archived (https://web.archive.org/web/20210817061036/https://www.reuters.com/article/us-iran-usa-stuxnet-idUSTRE81D24Q20120214) from the original on 17 August 2021. Retrieved 6 July 2021.
135. Beaumont, Peter (30 September 2010). "Stuxnet worm heralds new era of global cyberwar" (https://www.theguardian.com/technology/2010/sep/30/stuxnet-worm-new-era-global-cyberwar). *Guardian.co.uk*. London. Archived (https://web.archive.org/web/20161230195849/https://www.theguardian.com/technology/2010/sep/30/stuxnet-worm-new-era-global-cyberwar) from the original on 30 December 2016. Retrieved 17 December 2016.
136. Sanger, David E. (1 June 2012). "Obama Order Sped Up Wave of Cyberattacks Against Iran" (https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html?_r=2&pagewanted=2&seid=auto&smid=tw-nytimespolitics&pagewanted=all). *The New York Times*. Archived (https://web.archive.org/web/20220917203356/https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html?_r=2&pagewanted=2&seid=auto&smid=tw-nytimespolitics&pagewanted=all) from the original on 17 September 2022. Retrieved 1 June 2012.
137. Hounshell, Blake (27 September 2010). "6 mysteries about Stuxnet" (https://web.archive.org/web/20140209181744/http://blog.foreignpolicy.com/posts/2010/09/27/6_mysteries_about_stuxnet). *Foreign Policy*. Archived from the original (http://blog.foreignpolicy.com/posts/2010/09/27/6_mysteries_about_stuxnet) on 9 February 2014. Retrieved 28 September 2010.
138. "Falkenrath Says Stuxnet Virus May Have Origin in Israel: Video. Bloomberg Television" (https://web.archive.org/web/20121204121558/http://finance.yahoo.com/video/marketnews-19148628/falkenrath-says-stuxnet-virus-may-have-origin-in-israel-video-22090127#video=22147179). 24 September 2010. Archived from the original (https://finance.yahoo.com/video/marketnews-19148628/falkenrath-says-stuxnet-virus-may-have-origin-in-israel-video-22090127#video=22147179) on 4 December 2012.

139. Williams, Dan (15 December 2009). "Spymaster sees Israel as world cyberwar leader" (https://www.reuters.com/article/idUSTRE5BE30920091215). *Reuters*. Archived (https://web.archive.org/web/20101228170744/http://www.reuters.com/article/idUSTRE5BE30920091215) from the original on 28 December 2010. Retrieved 29 May 2012.
140. Dan Williams. "Cyber takes centre stage in Israel's war strategy" (https://web.archive.org/web/20101001091330/http://www.alertnet.org/thenews/newsdesk/LDE68R0GB.htm). Reuters, 28 September 2010. Archived from the original (http://www.alertnet.org/thenews/newsdesk/LDE68R0GB.htm) on 1 October 2010. Retrieved 17 October 2010.
141. Antonin Gregoire. "Stuxnet, the real face of cyber warfare" (http://www.iloubnan.info/politics/actualite/id/52568/titre/Stuxnet,-the-real-face-of-Cyber-Warfare). Iloubnan.info, 25 November 2010. Archived (https://web.archive.org/web/20101126171827/http://iloubnan.info/politics/actualite/id/52568/titre/Stuxnet,-the-real-face-of-Cyber-Warfare) from the original on 26 November 2010. Retrieved 25 November 2010.
142. Broad, William J.; Sanger, David E. (18 November 2010). "Worm in Iran Can Wreck Nuclear Centrifuges" (https://www.nytimes.com/2010/11/19/world/middleeast/19stuxnet.html). *The New York Times*. Archived (https://web.archive.org/web/20170219052914/http://www.nytimes.com/2010/11/19/world/middleeast/19stuxnet.html) from the original on 19 February 2017. Retrieved 25 February 2017.
143. Williams, Christopher (16 February 2011). "Israeli security chief celebrates Stuxnet cyber attack" (https://www.telegraph.co.uk/technology/news/8326274/Israeli-security-chief-celebrates-Stuxnet-cyber-attack.html). *The Telegraph*. London. Archived (https://web.archive.org/web/20110219063134/http://www.telegraph.co.uk/technology/news/8326274/Israeli-security-chief-celebrates-Stuxnet-cyber-attack.html) from the original on 19 February 2011. Retrieved 23 February 2011.
144. "The U.S. Cyber Consequences Unit" (http://www.usccu.us/). *The U.S. Cyber Consequences Unit*. Archived (https://web.archive.org/web/20230323151302/http://www.usccu.us/) from the original on 23 March 2023. Retrieved 1 December 2010.
145. "A worm in the centrifuge: An unusually sophisticated cyber-weapon is mysterious but important" (https://www.economist.com/international/2010/09/30/a-worm-in-the-centrifuge). *The Economist*. 30 September 2010. Archived (https://web.archive.org/web/20101010090003/http://www.economist.com/node/17147818) from the original on 10 October 2010. Retrieved 12 October 2010.
146. Sanger, David E. (25 September 2010). "Iran Fights Malware Attacking Computers" (https://www.nytimes.com/2010/09/26/world/middleeast/26iran.html?_r=2&src=me). *The New York Times*. Archived (https://web.archive.org/web/20110526234757/http://www.nytimes.com/2010/09/26/world/middleeast/26iran.html?_r=2&src=me) from the original on 26 May 2011. Retrieved 28 September 2010.
147. "Iran/Critical National Infrastructure: Cyber Security Experts See The Hand of Israel's Signals Intelligence Service in The 'Stuxnet' Virus Which Has Infected Iranian Nuclear Facilities" (https://web.archive.org/web/20101208204541/http://www.mideastsecurity.co.uk/?tag=stuxnet-myrtus-myrtle). Mideastsecurity.co.uk. 1 September 2010. Archived from the original (http://www.mideastsecurity.co.uk/?tag=stuxnet-myrtus-myrtle) on 8 December 2010. Retrieved 6 October 2010.
148. Riddle, Warren (1 October 2010). "Mysterious 'Myrtus' Biblical Reference Spotted in Stuxnet Code" (https://web.archive.org/web/20111001142357/http://www.switched.com/2010/10/01/mysterious-myrtus-biblical-reference-spotted-in-stuxnet-code/). SWITCHED. Archived from the original (http://www.switched.com/2010/10/01/mysterious-myrtus-biblical-reference-spotted-in-stuxnet-code/) on 1 October 2011. Retrieved 6 October 2010.
149. "SCADA Systems Whitepaper" (https://www.motorola.com/web/Business/Products/SCADA%20Products/_Documents/Static%20Files/SCADA_Sys_Wht_Ppr-2a_New.pdf) (PDF). Motorola. Archived (https://web.archive.org/web/20121001113541/http://www.motorola.com/web/Business/Products/SCADA%20Products/_Documents/Static%20Files/SCADA_Sys_Wht_Ppr-2a_New.pdf) (PDF) from the original on 1 October 2012. Retrieved 1 January 2016.

150. "Symantec Puts 'Stuxnet' Malware Under the Knife" (https://www.pcmag.com/article2/0,2817,2370107,00.asp). *PC Magazine*. Archived (https://web.archive.org/web/20170814140253/https://www.pcmag.com/article2/0,2817,2370107,00.asp) from the original on 14 August 2017. Retrieved 15 September 2017.
151. Zetter, Kim (1 October 2010). "New Clues Point to Israel as Author of Blockbuster Worm, Or Not" (https://www.wired.com/threatlevel/2010/10/stuxnet-deconstructed/). *Wired*. Archived (https://web.archive.org/web/20131215125816/http://www.wired.com/threatlevel/2010/10/stuxnet-deconstructed/) from the original on 15 December 2013. Retrieved 11 March 2017.
152. Reals, Tucker (24 September 2010). "Stuxnet Worm a U.S. Cyber-Attack on Iran Nukes?" (https://www.cbsnews.com/news/stuxnet-worm-a-us-cyber-attack-on-iran-nukes/). CBS News. Archived (https://web.archive.org/web/20131016133651/http://www.cbsnews.com/8301-501465_162-20017507-501465.html) from the original on 16 October 2013. Retrieved 27 September 2010.
153. "Snowden Der Spiegel Interview" (http://www.spiegel.de/international/world/interview-with-whistleblower-edward-snowden-on-global-spying-a-910006.html). *Der Spiegel* (in English and German). Archived (https://web.archive.org/web/20140706080155/http://www.spiegel.de/international/world/interview-with-whistleblower-edward-snowden-on-global-spying-a-910006.html) from the original on 6 July 2014. Retrieved 3 October 2015.
154. Kelley, Michael B. (1 June 2012). "Obama Administration Admits Cyberattacks Against Iran Are Part of Joint US-Israeli Offensive" (https://www.businessinsider.com/obama-cyberattacks-us-israeli-against-iran-2012-6). *Business Insider*. Archived (https://web.archive.org/web/20171203211236/http://www.businessinsider.com/obama-cyberattacks-us-israeli-against-iran-2012-6) from the original on 3 December 2017. Retrieved 23 January 2018.
155. Halliday, Josh (18 January 2011). "WikiLeaks: the US advised to sabotage Iran nuclear sites by German thinktank" (https://www.theguardian.com/world/2011/jan/18/wikileaks-us-embassy-cable-iran-nuclear). *The Guardian*. London. Archived (https://web.archive.org/web/20130908201914/http://www.theguardian.com/world/2011/jan/18/wikileaks-us-embassy-cable-iran-nuclear) from the original on 8 September 2013. Retrieved 19 January 2011.
156. Sanger, David E. (10 January 2009). "U.S. Rejected Aid for Israeli Raid on Iranian Nuclear Site" (https://www.nytimes.com/2009/01/11/washington/11iran.html?pagewanted=all&_r=0). *The New York Times*. Archived (https://web.archive.org/web/20131016065926/http://www.nytimes.com/2009/01/11/washington/11iran.html?pagewanted=all&_r=0) from the original on 16 October 2013. Retrieved 12 October 2013.
157. Kim Zetter (17 February 2011). "Cyberwar Issues Likely to Be Addressed Only After a Catastrophe" (https://www.wired.com/threatlevel/2011/02/cyberwar-issues-likely-to-be-addressed-only-after-a-catastrophe). *Wired*. Archived (https://web.archive.org/web/20110218154415/http://www.wired.com/threatlevel/2011/02/cyberwar-issues-likely-to-be-addressed-only-after-a-catastrophe/) from the original on 18 February 2011. Retrieved 18 February 2011.
158. Koop, Peter (12 December 2013). "Hoe onderschept de NSA ons dataverkeer?" (https://decorrespondent.nl/485/hoe-onderschept-de-nsa-ons-dataverkeer/31326229055-d9b3a5cf). *De Correspondent* (in Dutch). Archived (https://web.archive.org/web/20220222111223/https://decorrespondent.nl/485/hoe-onderschept-de-nsa-ons-dataverkeer/31326229055-d9b3a5cf) from the original on 22 February 2022. Retrieved 22 February 2022.
159. Chris Carroll (18 October 2011). "Cone of silence surrounds U.S. cyberwarfare" (http://www.stripes.com/news/cone-of-silence-surrounds-u-s-cyberwarfare-1.158090). *Stars and Stripes*. Archived (https://web.archive.org/web/20120307021747/http://www.stripes.com/news/cone-of-silence-surrounds-u-s-cyberwarfare-1.158090) from the original on 7 March 2012. Retrieved 30 October 2011.
160. John Bumgarner (27 April 2010). "Computers as Weapons of War" (https://web.archive.org/web/20111219174833/http://www.crows.org/images/stories/pdf/IOI/IO%20Journal_Vol2Iss2_0210.pdf) (PDF). IO Journal. Archived from the original (http://www.crows.org/images/stories/pdf/IOI/IO%20Journal_Vol2Iss2_0210.pdf) (PDF) on 19 December 2011. Retrieved 30 October 2011.

161. Kroft, Steve (4 March 2012). "Stuxnet: Computer worm opens new era of warfare" (https://web.archive.org/web/20131015171113/http://www.cbsnews.com/8301-18560_162-57390124/stuxnet-computer-worm-opens-new-era-of-warfare/). 60 Minutes (CBS News). Archived from the original (http://www.cbsnews.com/8301-18560_162-57390124/stuxnet-computer-worm-opens-new-era-of-warfare/) on 15 October 2013. Retrieved 9 March 2012.
162. CBS News staff (16 April 2011). "Iran blames U.S., Israel for Stuxnet malware" (https://www.cbsnews.com/news/iran-blames-us-israel-for-stuxnet-malware/) (SHTML). CBS News. Archived (https://web.archive.org/web/20120424151137/http://www.cbsnews.com/stories/2011/04/16/501364/main20054574.shtml) from the original on 24 April 2012. Retrieved 15 January 2012.
163. James Balford (12 June 2013). "The secret war" (https://www.wired.com/2013/06/general-keith-alexander-cyberwar/all/). *Wired*. Archived (https://web.archive.org/web/20180624045719/https://www.wired.com/2013/06/general-keith-alexander-cyberwar/all/) from the original on 24 June 2018. Retrieved 2 June 2014.
164. Carr, Jeffrey (14 December 2010). "Stuxnet's Finnish-Chinese Connection" (https://blogs.forbes.com/firewall/2010/12/14/stuxnets-finnish-chinese-connection/). *Forbes*. Archived (https://web.archive.org/web/20110418094630/http://blogs.forbes.com/firewall/2010/12/14/stuxnets-finnish-chinese-connection/) from the original on 18 April 2011. Retrieved 19 April 2011.
165. Clayton, Mark (24 September 2010). "Stuxnet worm mystery: What's the cyber weapon after?" (https://www.csmonitor.com/USA/2010/0924/Stuxnet-worm-mystery-What-s-the-cyber-weapon-after). *Christian Science Monitor*. Archived (https://web.archive.org/web/20100927170750/http://www.csmonitor.com/USA/2010/0924/Stuxnet-worm-mystery-What-s-the-cyber-weapon-after) from the original on 27 September 2010. Retrieved 21 January 2011.
166. "Revealed: How a secret Dutch mole aided the U.S.-Israeli Stuxnet cyberattack on Iran" (https://news.yahoo.com/revealed-how-a-secret-dutch-mole-aided-the-us-israeli-stuxnet-cyber-attack-on-iran-160026018.html). *news.yahoo.com*. 2 September 2019. Archived (https://web.archive.org/web/20190903172309/https://news.yahoo.com/revealed-how-a-secret-dutch-mole-aided-the-us-israeli-stuxnet-cyber-attack-on-iran-160026018.html) from the original on 3 September 2019. Retrieved 3 September 2019.
167. Bob, Yonah Jeremy (2 September 2019). "Secret Dutch mole aided Stuxnet attack on Iran's nuke program – Report" (https://www.jpost.com/Middle-East/Secret-Dutch-mole-aided-Stuxnet-attack-on-Irans-nuke-program-report-600430). *Jerusalem Post*. Archived (https://web.archive.org/web/20190905010321/https://www.jpost.com/Middle-East/Secret-Dutch-mole-aided-Stuxnet-attack-on-Irans-nuke-program-report-600430) from the original on 5 September 2019. Retrieved 4 September 2019.
168. Gaycken, Sandro (26 November 2010). "Stuxnet: Wer war's? Und wozu?" (http://www.zeit.de/2010/48/Computerwurm-Stuxnet). *Die ZEIT*. Archived (https://web.archive.org/web/20110420065905/http://www.zeit.de/2010/48/Computerwurm-Stuxnet) from the original on 20 April 2011. Retrieved 19 April 2011.
169. Hopkins, Nick (31 May 2011). "UK developing cyber-weapons programme to counter cyber war threat" (https://www.theguardian.com/uk/2011/may/30/military-cyberwar-offensive). *The Guardian*. United Kingdom. Archived (https://web.archive.org/web/20130910083707/http://www.theguardian.com/uk/2011/may/30/military-cyberwar-offensive) from the original on 10 September 2013. Retrieved 31 May 2011.
170. Iain Thomson (8 July 2013). "Snowden: US and Israel Did Create Stuxnet Attack Code" (https://www.theregister.co.uk/2013/07/08/snowden_us_israel_stuxnet/). *The Register*. Archived (https://web.archive.org/web/20130710195741/http://www.theregister.co.uk/2013/07/08/snowden_us_israel_stuxnet/) from the original on 10 July 2013. Retrieved 8 July 2013.
171. Menn, Joseph (29 May 2015). "Exclusive: U.S. tried Stuxnet-style campaign against North Korea but failed – sources" (https://www.reuters.com/article/us-usa-northkorea-stuxnet-idUSKBN0OE2DM20150529). *Reuters*. Archived (https://web.archive.org/web/20151213161547/http://www.reuters.com/article/us-usa-northkorea-stuxnet-idUSKBN0OE2DM20150529) from the original on 13 December 2015. Retrieved 31 May 2015.

172. Goud, Naveen (6 November 2018). "Iran says Israel launched Stuxnet 2.0 Cyber Attack" (https://www.cybersecurity-insiders.com/iran-says-israel-launched-stuxnet-2-0-cyber-attack/). Archived (https://web.archive.org/web/20190207021512/https://www.cybersecurity-insiders.com/iran-says-israel-launched-stuxnet-2-0-cyber-attack/) from the original on 7 February 2019. Retrieved 6 February 2019.
173. "Stuxnet's Secret Twin" (https://foreignpolicy.com/articles/2013/11/19/stuxnets_secret_twin_iran_nukes_cyber_attack). *Foreign Policy*. 19 November 2013. Archived (https://web.archive.org/web/20141204132954/http://www.foreignpolicy.com/articles/2013/11/19/stuxnets_secret_twin_iran_nukes_cyber_attack) from the original on 4 December 2014. Retrieved 11 March 2017.
174. "Duqu: A Stuxnet-like malware found in the wild, technical report" (http://www.crysys.hu/publications/files/bencsathPBF11duqu.pdf) (PDF). Laboratory of Cryptography of Systems Security (CrySyS). 14 October 2011. Archived (https://web.archive.org/web/20190421025710/https://www.crysys.hu/publications/files/bencsathPBF11duqu.pdf) (PDF) from the original on 21 April 2019. Retrieved 13 November 2011.
175. "Statement on Duqu's initial analysis" (https://web.archive.org/web/20121004111047/http://crysys.hu/in-the-press.html). Laboratory of Cryptography of Systems Security (CrySyS). 21 October 2011. Archived from the original (http://www.crysys.hu/in-the-press.html) on 4 October 2012. Retrieved 25 October 2011.
176. "W32.Duqu – The precursor to the next Stuxnet (Version 1.2)" (https://web.archive.org/web/20191025145245/https://www.symantec.com/content/en/us/enterprise/media/security_response/whitepapers/w32_duqu_the_precursor_to_the_next_stuxnet.pdf) (PDF). Symantec. 20 October 2011. Archived from the original (http://www.symantec.com/content/en/us/enterprise/media/security_response/whitepapers/w32_duqu_the_precursor_to_the_next_stuxnet.pdf) (PDF) on 25 October 2019. Retrieved 25 October 2011.
177. Finkle, Jim (28 December 2011). "Stuxnet weapon has at least 4 cousins: researchers" (https://www.reuters.com/article/us-cybersecurity-stuxnet-idUSTRE7BR1EV20111228). *Reuters*. Archived (https://web.archive.org/web/20150924161301/http://www.reuters.com/article/2011/12/28/us-cybersecurity-stuxnet-idUSTRE7BR1EV20111228) from the original on 24 September 2015. Retrieved 6 July 2021.
178. Zetter, Kim (28 May 2012). "Meet 'Flame,' The Massive Spy Malware Infiltrating Iranian Computers" (https://www.wired.com/threatlevel/2012/05/flame/). *Wired*. Archived (https://web.archive.org/web/20120530213153/http://www.wired.com/threatlevel/2012/05/flame/) from the original on 30 May 2012. Retrieved 29 May 2012.
179. "Resource 207: Kaspersky Lab Research Proves that Stuxnet and Flame Developers are Connected" (http://www.kaspersky.com/about/news/virus/2012/Resource_207_Kaspersky_Lab_Research_Proves_that_Stuxnet_and_Flame_Developers_are_Connected). Kaspersky Lab. 11 June 2012. Archived (https://web.archive.org/web/20211116092936/https://www.kaspersky.com/about/press-releases) from the original on 16 November 2021. Retrieved 13 June 2012.
180. "The Meaning of Stuxnet" (https://www.economist.com/leaders/2010/09/30/the-meaning-of-stuxnet). *The Economist*. 30 September 2010. Archived (https://web.archive.org/web/20150330224907/http://www.economist.com/node/17147862) from the original on 30 March 2015. Retrieved 18 April 2015.
181. Kim Zetter (8 July 2011). "How Digital Detectives Deciphered Stuxnet, the Most Menacing Malware in History" (https://www.wired.com/2011/07/how-digital-detectives-deciphered-stuxnet/). *Wired*. Archived (https://web.archive.org/web/20170309033011/https://www.wired.com/2011/07/how-digital-detectives-deciphered-stuxnet/) from the original on 9 March 2017. Retrieved 11 March 2017.

182. Holger Stark (8 August 2011). "Mossad's Miracle Weapon: Stuxnet Virus Opens New Era of Cyber War" (http://www.spiegel.de/international/world/mossad-s-miracle-weapon-stuxnet-virus-opens-new-era-of-cyber-war-a-778912.html). *Der Spiegel*. Archived (https://web.archive.org/web/20150412223510/http://www.spiegel.de/international/world/mossad-s-miracle-weapon-stuxnet-virus-opens-new-era-of-cyber-war-a-778912.html) from the original on 12 April 2015. Retrieved 18 April 2015.
183. Eddie Walsh (1 January 2012). "2011: The year of domestic cyber threat" (http://www.aljazeera.com/indepth/opinion/2012/01/20121181245291117.html). Al Jazeera English. Archived (https://web.archive.org/web/20150418100836/http://www.aljazeera.com/indepth/opinion/2012/01/20121181245291117.html) from the original on 18 April 2015. Retrieved 18 April 2015.
184. Vyacheslav Zakorzhevsky (5 October 2010). "Sality & Stuxnet – Not Such a Strange Coincidence" (http://www.kaspersky.com/about/news/virus/2010/Sality_Stuxnet_Not_Such_a_Strange_Coincidence). Kaspersky Lab. Archived (https://web.archive.org/web/20150418101908/http://www.kaspersky.com/about/news/virus/2010/Sality_Stuxnet_Not_Such_a_Strange_Coincidence) from the original on 18 April 2015. Retrieved 18 April 2015.
185. Ball, James (16 February 2016). "U.S. Hacked into Iran's Critical Civilian Infrastructure For Massive Cyberattack, New Film Claims" (https://www.buzzfeed.com/jamesball/us-hacked-into-irans-critical-civilian-infrastructure-for-ma). *BuzzFeed*. Archived (https://web.archive.org/web/20170719204217/https://www.buzzfeed.com/jamesball/us-hacked-into-irans-critical-civilian-infrastructure-for-ma) from the original on 19 July 2017. Retrieved 17 May 2017.
186. Savage, Charlie (17 October 2016). "James Cartwright, Ex-General, Pleads Guilty in Leak Case" (https://www.nytimes.com/2016/10/18/us/marine-general-james-cartwright-leak-fbi.html). *The New York Times*. ISSN 0362-4331 (https://search.worldcat.org/issn/0362-4331). Archived (https://web.archive.org/web/20170112112828/https://www.nytimes.com/2016/10/18/us/marine-general-james-cartwright-leak-fbi.html) from the original on 12 January 2017. Retrieved 27 December 2016.
187. "World War Three, by Mistake" (https://www.newyorker.com/news/news-desk/world-war-three-by-mistake). *The New Yorker*. 23 December 2016. Archived (https://web.archive.org/web/20161227001300/http://www.newyorker.com/news/news-desk/world-war-three-by-mistake) from the original on 27 December 2016. Retrieved 27 December 2016.
188. "Stuxnet – Darknet Diaries" (https://darknetdiaries.com/episode/29/). *Darknet Diaries – True stories from the dark side of the Internet*. 2 January 2019. Retrieved 11 October 2025.
189. "Splinter Cell Blacklist – Mission 10 'American Fuel'" (https://www.youtube.com/watch?v=o77cNhuoI4A). 17 September 2013. Archived (https://ghostarchive.org/varchive/youtube/20211221/o77cNhuoI4A) from the original on 21 December 2021 – via www.youtube.com.
190. "According to Star Trek: Discovery, Starfleet still runs Microsoft Windows" (https://www.theverge.com/tldr/2017/10/3/16412372/star-trek-discovery-cbs-windows-code-command-line). *The Verge*. 3 October 2017. Archived (https://web.archive.org/web/20190111121502/https://www.theverge.com/tldr/2017/10/3/16412372/star-trek-discovery-cbs-windows-code-command-line) from the original on 11 January 2019. Retrieved 11 January 2019.

Further reading

- Langner, Ralph (March 2011). "Ralph Langner: Cracking Stuxnet, a 21st-century cyber weapon" (https://www.ted.com/talks/ralph_langner_cracking_stuxnet_a_21st_century_cyber_weapon). TED. Archived (https://web.archive.org/web/20140201101050/http://www.ted.com/talks/ralph_langner_cracking_stuxnet_a_21st_century_cyberweapon.html) from the original on 1 February 2014. Retrieved 13 May 2011.
- "The short path from cyber missiles to dirty digital bombs" (http://www.langner.com/en/2010/12/26/the-short-path-from-cyber-missiles-to-dirty-digital-bombs/). Blog. Langner Communications GmbH. 26 December 2010. Archived (https://web.archive.org/web/201704

19094857/http://www.langner.com/en/2010/12/26/the-short-path-from-cyber-missiles-to-dirty-digital-bombs/) from the original on 19 April 2017. Retrieved 13 May 2011.

- Ralph Langner's Stuxnet Deep Dive (http://www.digitalbond.com/2012/01/31/langners-stuxnet-deep-dive-s4-video/) Archived (https://web.archive.org/web/20121017163356/https://www.digitalbond.com/2012/01/31/langners-stuxnet-deep-dive-s4-video/) 17 October 2012 at the Wayback Machine
- Langner, Ralph (November 2013). To Kill a Centrifuge: A Technical Analysis of What Stuxnet's Creators Tried to Achieve (http://www.langner.com/en/wp-content/uploads/2013/11/To-kill-a-centrifuge.pdf) (PDF) (Report). Archived (https://web.archive.org/web/20160613132728/http://www.langner.com/en/wp-content/uploads/2013/11/To-kill-a-centrifuge.pdf) (PDF) from the original on 13 June 2016. Retrieved 26 November 2013.
- Falliere, Nicolas (21 September 2010). "Exploring Stuxnet's PLC Infection Process" (https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=ad4b3d10-b808-414c-b4c3-ae4a2ed85560&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments). Blogs: Security Response. Symantec. Archived (https://web.archive.org/web/20210621221438/https://community.broadcom.com/symantecenterprise/communities/community-home/librarydocuments/viewdocument?DocumentKey=ad4b3d10-b808-414c-b4c3-ae4a2ed85560&CommunityKey=1ecf5f55-9545-44d6-b0f4-4e4a7f5f5e68&tab=librarydocuments) from the original on 21 June 2021. Retrieved 13 May 2011.
- "Stuxnet Questions and Answers" (http://www.f-secure.com/weblog/archives/00002040.html). News from the Lab (blog). F-Secure. 1 October 2010. Archived (https://web.archive.org/web/20210505195313/https://archive.f-secure.com/weblog/archives/00002040.html) from the original on 5 May 2021. Retrieved 13 May 2011.
- Dang, Bruce; Ferrie, Peter (28 December 2010). "27C3: Adventures in analyzing Stuxnet" (http://media.ccc.de/browse/congress/2010/27c3-4245-en-adventures_in_analyzing_stuxnet.html). Chaos Computer Club e.V. Archived (https://web.archive.org/web/20151011213142/http://media.ccc.de/browse/congress/2010/27c3-4245-en-adventures_in_analyzing_stuxnet.html) from the original on 11 October 2015. Retrieved 13 May 2011.
- Russinovich, Mark (30 March 2011). "Analyzing a Stuxnet Infection with the Sysinternals Tools, Part 1" (https://web.archive.org/web/20110423024408/http://blogs.technet.com/b/markrussinovich/archive/2011/03/30/3416253.aspx). Mark's Blog. Microsoft Corporation. MSDN Blogs. Archived from the original (http://blogs.technet.com/b/markrussinovich/archive/2011/03/30/3416253.aspx) on 23 April 2011. Retrieved 13 May 2011.
- Zetter, Kim (11 July 2011). "How Digital Detectives Deciphered Stuxnet, the Most Menacing Malware in History" (https://www.wired.com/threatlevel/2011/07/how-digital-detectives-deciphered-stuxnet/all/1). Threat Level Blog. *Wired*. Archived (https://web.archive.org/web/20140328025407/http://www.wired.com/threatlevel/2011/07/how-digital-detectives-deciphered-stuxnet/all/1) from the original on 28 March 2014. Retrieved 11 July 2011.
- Kroft, Steve (4 March 2012). "Stuxnet: Computer worm opens new era of warfare" (https://web.archive.org/web/20131015171113/http://www.cbsnews.com/8301-18560_162-57390124/stuxnet-computer-worm-opens-new-era-of-warfare/). *60 Minutes*. CBS News. Archived from the original (http://www.cbsnews.com/8301-18560_162-57390124/stuxnet-computer-worm-opens-new-era-of-warfare/) on 15 October 2013. Retrieved 4 March 2012.
- Sanger, David E. (1 June 2012). "Obama Order Sped Up Wave of Cyberattacks Against Iran" (https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html?_r=2&pagewanted=2&seid=auto&smid=tw-nytimespolitics&pagewanted=all). *The New York Times*. Archived (https://web.archive.org/web/20220917203356/https://www.nytimes.com/2012/06/01/world/middleeast/obama-ordered-wave-of-cyberattacks-against-iran.html?_r=2&pagewanted=2&seid=auto&smid=tw-nytimespolitics&pagewanted=all) from the original on 17 September 2022. Retrieved 1 June 2012.
- Kim Zetter, *Countdown to Zero Day: Stuxnet and the Launch of the World's First Digital Weapon*. New York: Crown Publishing Group, 2014. ISBN 978-0-7704-3617-9.

External links

- fanny.bmp (https://securelist.com/a-fanny-equation-i-am-your-father-stuxnet/68787) – at Securelist
- fanny.bmp source (https://github.com/loneicewolf/fanny.bmp) – at GitHub
- Stuxnet code (https://archive.org/details/Stuxnet) – at Internet Archive

Retrieved from "https://en.wikipedia.org/w/index.php?title=Stuxnet&oldid=1345628122"